Exact Reference-Assisted Entropy Contraction from the Choi Matrix

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14 November 2026

Abstract

How much distinguishability can a quantum channel preserve when its input is correlated with an untouched quantum reference? We give an exact finite semidefinite program for the complete, equal-reference Umegaki contraction coefficient of every finite-dimensional channel. Its value also equals the reference-subtracted relative-entropy coefficient and the conditional mutual-information coefficient. The program uses the unnormalized Choi matrix; its inverse formulation requires an explicit range condition. The proof identifies a centered complete Gram optimization, constructs a state-independent signed observable lift through an erasure channel, and proves sharpness with a rare coherent reference sector. References of bounded dimension suffice, even when both input marginals are prescribed and faithful. The same program computes the corresponding coefficients for regular modular Petz divergences, each fixed Petz Rényi order between zero and one, and quantum Chernoff divergence. Consequences include a harmonic dual, exact tensor products, a sharp comparison with positive CP-replacer minorization, and explicit classical and unitary-noise formulas. We separately compute ordinary quantum mutual-information contraction for qubit depolarization, which is strictly smaller. A discontinuity at pure reset rules out uniformly bounded-use black-box estimation, distinguishing exact Choi-data evaluation from robust acquisition of an unknown channel.

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1 Introduction

A noisy channel reduces the distinguishability of quantum states. Quantifying that reduction is a basic problem in information theory, with uses in statistical comparison, information propagation and the analysis of noisy dynamics. The answer depends on what information is allowed outside the channel. An untouched reference can retain information perfectly. Consequently, the unrestricted ratio of input and output relative entropies becomes trivial if the compared reference states may differ and that contribution is left unaccounted for.

Two established definitions avoid this problem. The *complete* coefficient compares joint states with the same reference marginal. The *conditional* coefficient instead subtracts the reference relative entropy from both numerator and denominator. George, Hirche, Nuradha and Wilde introduce the complete coefficient and CP-replacer bounds in their quantum Doeblin framework [6, Definition 11 and Section 6.1]. Hirche, George, Nuradha and Wilde develop the conditional framework and identify its Umegaki coefficient with conditional-mutual-information contraction [9, Theorem 1 and Corollary 3]. Their Section 3.3 asks how complete, conditional and ordinary mutual-information contraction compare, including for depolarizing channels.

The main contribution here is an exact value for the complete and conditional coefficients of an arbitrary finite channel. An optimization over references and state pairs of unbounded dimension is replaced by one finite semidefinite program in the channel's Choi matrix. A matching construction proves that the program gives the coefficient itself, rather than a sufficient upper bound. It also bounds the reference dimension needed to approach the optimum. The complete and conditional Umegaki coefficients coincide for every channel. For qubit depolarization, the ordinary quantum mutual-information coefficient is strictly smaller, as shown by a separate argument.

Foundations and attribution. The proof uses established machinery in specific roles. Stinespring factorization [16], Choi's representation [3] and convex minimax [15] provide the finite geometry. Signed observable lifting and the conversion of complete weighted-square inequalities into relative-entropy comparison are developed by Zhu and Wang [17, Definition S3, Proposition S4, Theorem S5 and Lemmas S2, S6–S7]. Hirche establishes related complete estimation orders and signed-lift characterizations [8, Theorem 3.2 and Proposition 3.5]. We prove the particular lifting and integration statements needed here. The additional operations are solving the centered one-channel optimization, reducing its complex center to a positive one, and obtaining its sharp coherent converse.

The modular-divergence extension uses the positive integral representation of operator-convex functions [4, Theorem 2.2]. It concerns modular Petz divergences, unlike the hockey-stick integral divergences of Hirche and Tomamichel [10]. Ordinary, unreferenced contraction coefficients can depend on the chosen divergence [7]. Complete entropy decay toward a specified equilibrium or conditional expectation [5] is also a different optimization. These distinctions are material, not changes of notation.

Consequences and limits. The formula gives a common-output-row criterion for strict contraction, a harmonic dual, an exact product law and prescribed-input-marginal refinements. We derive exact classical, depolarizing and orthogonal-unitary-noise formulas. For the regular modular Petz class, each fixed Petz Rényi order $0 < \alpha < 1$, and quantum Chernoff divergence, the same finite program gives the corresponding reference-assisted coefficients. The Chernoff interpretation uses the established iid binary-testing theorem [1, 12]; it is not a solution of a multi-state quantum exclusion problem.

Exact Choi data are a strong input contract. A Choi matrix is dense in Hilbert-space dimension and can be exponentially large in the number of physical qubits. Moreover, the coefficient is

Established ingredient	Role retained here	Added conclusion
Complete and conditional coefficient frameworks [6, 9]	Definitions, reference subtraction and information interpretation	Exact common all-channel value and bounded-reference sharpness
Signed lifts and metric comparison [17, 8]	Statistical reconstruction without positivity of the lift	Optimal comparison with erasure, expressed in the Choi matrix
Positive-resolvent representation [4]	Integral decomposition of regular operator-convex generators	One exact coefficient for the stated modular Petz class and logarithmic extensions
Classical SDPI and mixing theory [13, 11]	Ordinary comparison methods and tensor-product cutoff context	Explicit reference-assisted values and a scoped Doeblin comparison

Table 1: The main construction is the exact Choi optimization and converse. The later examples and applications use this construction and established foundations.

discontinuous at pure reset. We prove that no uniformly bounded number of black-box channel uses estimates it to additive error below one half on all channels. Rare near-extremizers can carry vanishing absolute information. Neither a large contraction ratio nor a signed lift therefore implies a communication rate, a physical simulator or a recovery map.

1.1 Conceptual overview

The proof has two directions that meet at the same finite quadratic problem. For the upper bound, ask how far every output observable is from a common scalar center after applying the adjoint channel. A complete Gram inequality records these deviations and all their cross terms at once. Stinespring factorization turns that inequality into a norm minimization. Vectorizing its Kraus operators produces the inverse-Choi objective. A polar-decomposition argument then shows that the center can be chosen positive.

That positive center defines a signed observable map through an erasure channel: erasure either retains the input or records only its reference marginal. The map reconstructs expectations and controls both orders of multiplication in weighted squares. Inverse quadratic duality converts these inequalities into local entropy metric comparison. Integrating along an affine path of states gives the global reference-subtracted inequality. The observable map need not preserve positive operators; no physical postprocessing is used.

For sharpness, the dual quadratic problem is rectangular. Embed a trace-zero rectangular perturbation as coherence between an ordinary reference sector and a rare additional sector. Both local marginals remain *exactly* unchanged. First take the perturbation to zero at fixed rare-sector weight. Only afterward send that weight to zero. The resulting entropy ratio is the rectangular quadratic ratio, which is already equal to the Gram optimum. The construction needs only finitely many reference dimensions.

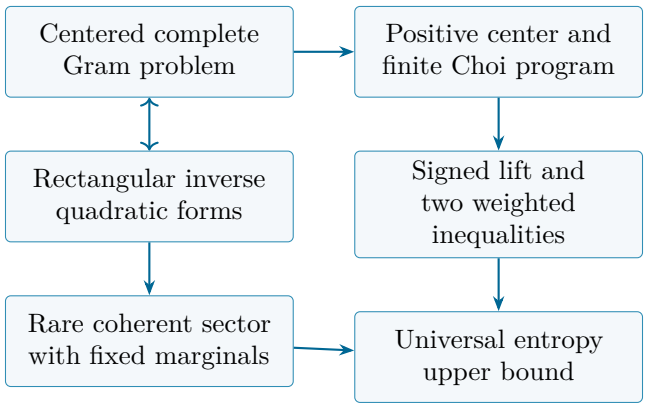


Figure 1: The upper inequality and the coherent converse meet at an exact finite optimization.

The remainder follows this order. Sections 3–5 prove the main theorem. Section 6 develops its geometry and products. Sections 7–8 prove the divergence extensions. Exact families, noise applications and the acquisition obstruction occupy Sections 9–11.

2 Coefficients and the exact Choi formula

All Hilbert spaces are finite dimensional. Write $\mathcal{L}(A)$ for the operators on A and $\mathsf{D}(A)$ for its density operators. Let $\mathcal{N} : \mathcal{L}(A) \to \mathcal{L}(B)$ be completely positive and trace preserving (CPTP), and put $d = \dim A$, $m = \dim B$. We use natural logarithms and the Hilbert–Schmidt inner product $\langle X, Y\rangle = \operatorname{Tr} X^* Y$. A superscript $*$ on a map denotes its Hilbert–Schmidt adjoint. Reference extensions are denoted $\mathcal{N}_R = \mathrm{id}_R \otimes \mathcal{N}$.

The Umegaki relative entropy is

$$D(\rho\|\sigma) = \begin{cases} \operatorname{Tr}\rho(\log\rho - \log\sigma), & \operatorname{supp}\rho \subseteq \operatorname{supp}\sigma, \\ +\infty, & \text{otherwise.} \end{cases}$$

Here $0\log 0 = 0$, and logarithms on supports have their usual spectral meaning. Unless explicitly stated otherwise, every ratio below has a strictly positive, finite denominator. In a difference of divergences, each term must be finite; an expression $\infty - \infty$ is never allowed. An empty supremum is zero.

Definition 2.1 (Three reference-assisted coefficients)**.** The complete and conditional-subtraction coefficients are

$$\eta_D^{\mathrm{c}}(\mathcal{N}) = \sup_{\substack{R,\rho,\sigma\in\mathsf{D}(RA)\\ \rho_R=\sigma_R,\ 0<D(\rho\|\sigma)<\infty}} \frac{D(\mathcal{N}_R\rho\|\mathcal{N}_R\sigma)}{D(\rho\|\sigma)}, \tag{2.1}$$

$$\eta_D^{\Delta}(\mathcal{N}) = \sup_{\substack{R,\rho,\sigma\in\mathsf{D}(RA)\\ D(\rho\|\sigma)<\infty,\ D(\rho\|\sigma)>D(\rho_R\|\sigma_R)}} \frac{D(\mathcal{N}_R\rho\|\mathcal{N}_R\sigma) - D(\rho_R\|\sigma_R)}{D(\rho\|\sigma) - D(\rho_R\|\sigma_R)}. \tag{2.2}$$

The conditional-mutual-information coefficient is

$$\eta_{\mathrm{condMI}}(\mathcal{N}) = \sup_{\substack{U,R,\rho\in\mathsf{D}(URA)\\ I(U;A|R)_\rho>0}} \frac{I(U;B|R)_{(\mathrm{id}_{UR}\otimes\mathcal{N})(\rho)}}{I(U;A|R)_\rho}. \tag{2.3}$$

Both U and R may be quantum. Here $I(U;A|R) = S(UR) + S(AR) - S(R) - S(URA)$ and $S(\rho) = -\operatorname{Tr}\rho\log\rho$.

The coefficient in (2.3) has an additional conditioning system. It differs from ordinary quantum mutual-information contraction,

$$\eta_{\mathrm{MI}}(\mathcal{N}) = \sup_{\substack{R,\rho\in\mathsf{D}(RA)\\ I(R;A)_\rho>0}} \frac{I(R;B)_{\mathcal{N}_R(\rho)}}{I(R;A)_\rho}. \tag{2.4}$$

The reference R in (2.4) is still arbitrary and quantum. The adjective *ordinary* does not mean that this reference is classical.

We use the *unnormalized* Choi operator

$$J_{\mathcal{N}} = \sum_{i,j=1}^{d} |i\rangle\langle j| \otimes \mathcal{N}(|i\rangle\langle j|), \qquad \operatorname{Tr}_B J_{\mathcal{N}} = I_A. \tag{2.5}$$

Write $J = J_{\mathcal{N}}$. Its Moore–Penrose inverse is J^+.

Notation	Optimization	Reference requirement
η_D^{ord}	$D(\mathcal{N}\rho\|\mathcal{N}\sigma)/D(\rho\|\sigma)$	No reference
η_{MI}	$I(R;B)/I(R;A)$	One arbitrary quantum reference
η_D^{prod}	$D(\mathcal{N}_R\rho\|\rho_R\otimes\mathcal{N}\sigma_A)/D(\rho\|\rho_R\otimes\sigma_A)$	Second joint state is a product
η_D^{c}	(2.1)	Compared reference marginals equal
η_D^{Δ}	(2.2)	Reference divergence subtracted
η_{condMI}	(2.3)	Separate message U and memory R

Table 2: The optimizations are different. The product-reference coefficient also optimizes σ_A; fixing it gives a further, generally different, quantity.

Definition 2.2 (The Choi optimization)**.** Let $M(\mathcal{N})$ be the value of the finite SDP

$$\begin{aligned} \text{minimize} \quad & t \\ \text{over} \quad & \sigma \in \mathsf{D}(B), \quad Z = Z^* \in \mathcal{L}(AB), \quad t \in \mathbb{R} \\ \text{subject to} \quad & \begin{pmatrix} J & I_A \otimes \sigma \\ I_A \otimes \sigma & Z \end{pmatrix} \succeq 0, \qquad \mathrm{Tr}_B\, Z \preceq t I_A. \end{aligned} \tag{2.6}$$

If it is infeasible, set $M(\mathcal{N}) = +\infty$. Set $\beta(\mathcal{N}) = 1/M(\mathcal{N})$, with $1/\infty = 0$. The output density σ in this program is an optimization center, independent of the compared state pairs in (2.1).

Theorem 2.3 (Exact reference-assisted Umegaki contraction)**.** *For every finite-dimensional CPTP map,*

$$\boxed{\eta_D^{\mathrm{c}}(\mathcal{N}) = \eta_D^{\Delta}(\mathcal{N}) = \eta_{\mathrm{condMI}}(\mathcal{N}) = 1 - \beta(\mathcal{N}).} \tag{2.7}$$

Whenever (2.6) *is feasible its minimum is attained and* $M \geq 1$*. Equivalently,*

$$M = \min_{\substack{\sigma \in \mathsf{D}(B) \\ \mathrm{ran}(I_A \otimes \sigma) \subseteq \mathrm{ran}\, J}} \|\mathrm{Tr}_B[(I_A \otimes \sigma) J^+ (I_A \otimes \sigma)]\|_\infty. \tag{2.8}$$

For every finite R *and state pair with* $D(\rho\|\sigma) < \infty$*,*

$$D(\mathcal{N}_R\rho\|\mathcal{N}_R\sigma) \leq (1-\beta) D(\rho\|\sigma) + \beta D(\rho_R\|\sigma_R). \tag{2.9}$$

The supremum in (2.1) *can be approached with* $\dim R \leq m^2 + 1$ *and with both local marginals unchanged. In* (2.3)*, a uniform classical bit* U *suffices, with* $\dim R \leq m^2 + 1$ *and* $I(U;R) = I(U;A) = 0$ *exactly. If* $d = 1$*, all three coefficients are zero.*

The range condition in (2.8) is part of the formula. For example, the identity channel on a space of dimension greater than one has rank-one Choi operator and no feasible density σ. It therefore has $M = \infty$ and coefficient one. Substituting a pseudoinverse while dropping the range constraint would incorrectly make a finite expression determine this case. Using a normalized Choi state would also change the scale of the optimization.

The definition-level distinction between conditional and ordinary mutual information can now be summarized precisely.

Proposition 2.4 (The coefficient hierarchy)**.** *With the quantities in table 2,*

$$0 \le \eta_D^{\mathrm{ord}}(\mathcal{N}) \le \eta_{\mathrm{MI}}(\mathcal{N}) = \eta_D^{\mathrm{prod}}(\mathcal{N}) \le \eta_D^{\mathrm{c}}(\mathcal{N}) = \eta_D^{\Delta}(\mathcal{N}) = \eta_{\mathrm{condMI}}(\mathcal{N}) \le 1. \tag{2.10}$$

The identity $\eta_{\mathrm{MI}} = \eta_D^{\mathrm{prod}}$ and the underlying information correspondences are established in [9, Corollary 3]. A proof of the product-reference identity is included in appendix B.1; the complete/conditional equalities are proved below.

Thus equality in (2.7) does not identify ordinary MI with conditional MI. For qubit depolarization at shrinkage $1/2$, the three successive values $\eta_D^{\mathrm{ord}}, \eta_{\mathrm{MI}}, \eta_D^{\mathrm{c}}$ are $1/4, 1/3, 3/8$; see theorem 9.4. Fixing both A marginals to a faithful state also does not fix the second joint state to a product. The former restriction leaves (2.7) unchanged, as proved in theorem 6.5.

3 A finite Gram problem and its Choi solution

The finite optimization enters before entropy. We first measure how far the adjoint channel's output observables are from a common scalar center. Keeping the whole Gram matrix is essential: its off-diagonal entries retain the tests that arise from quantum references.

3.1 Output supports and centered Gram matrices

Let $B_{\mathrm{s}} = \operatorname{supp}\mathcal{N}(I_A)$. Every channel output is supported on B_{s}, since $X \succeq 0$ implies $X \preceq cI_A$ for some c. The same holds after every reference extension. If $C \succ 0$ on RA, then

$$\mathcal{N}_R(C) \succeq c\, I_R \otimes \mathcal{N}(I_A) \succ 0 \quad \text{on } R \otimes B_{\mathrm{s}}$$

for some $c > 0$. We may therefore work on this fixed output space. A feasible center in (2.8) must be supported there, so the optimization is also unchanged by this reduction. Inverses of faithful output states below are always taken on this actual output space. For notational ease call its dimension m in the proof; it is at most the original m.

Put $\Phi = \mathcal{N}^*$, a unital completely positive map. Choose a complex basis $F_1, \dots, F_n$ of $\mathcal{L}(B)$, where $n = m^2$, and define

$$\mathbf{A} = \begin{pmatrix} \Phi(F_1) \\ \vdots \\ \Phi(F_n) \end{pmatrix}, \qquad \mathbf{G} = [\Phi(F_i F_j^*)]_{i,j=1}^n.$$

Let η_{G} be the least $\eta \ge 0$ for which some $a \in \mathbb{C}^n$ obeys

$$(\mathbf{A} - a \otimes I_A)(\mathbf{A} - a \otimes I_A)^* \preceq \eta \mathbf{G}. \tag{3.1}$$

Complete Schwarz gives feasibility at $\eta = 1, a = 0$. A bounded value of η bounds $\|\mathbf{A} - a \otimes I\|$ and hence $\|a\|$, so a minimizing center exists. A different finite tuple of output operators is a scalar linear combination of this basis. Congruence of (3.1) therefore controls every such tuple with the same linear centering functional.

3.2 The rectangular dual problem

For $C \succ 0$ on $\mathbb{C}^k \otimes A$, a block column $B : A \to \mathbb{C}^k \otimes A$ consists of k matrices $B_i \in \mathcal{L}(A)$. Define

$$q_C(B) = \operatorname{Tr} B^* C^{-1} B, \qquad \operatorname{Tr}_A B = (\operatorname{Tr} B_1, \dots, \operatorname{Tr} B_k)^{\mathsf{T}}.$$

Write $C' = \mathcal{N}_{\mathbb{C}^k}(C)$ and $B' = (\mathcal{N}(B_i))_i$. The condition $\operatorname{Tr}_A B = 0$ is blockwise; it is stronger than a single total trace constraint.

Lemma 3.1 (Complete Gram–rectangular identity)**.** *Let*

$$\eta_{\mathrm{rect}} = \sup_{\substack{k,\ C\succ 0,\ B\neq 0 \\ \operatorname{Tr}_A B = 0}} \frac{q_{C'}(B')}{q_C(B)}. \tag{3.2}$$

Then $\eta_{\mathrm{rect}} = \eta_{\mathrm{G}}$*. The supremum is unchanged by requiring* $\operatorname{Tr} C = 1$ *and* $k \le m^2$*.*

Proof. We give the weighted duality and the minimax step separately. For an output block column $F = (F_i)_i$, Hilbert-space projection gives

$$\sup_{\substack{B\neq 0 \\ \operatorname{Tr}_A B = 0}} \frac{|\operatorname{Tr} B^* \Phi(F)|^2}{q_C(B)} = \inf_{a \in \mathbb{C}^k} \operatorname{Tr} C\,(\Phi(F) - a \otimes I)(\Phi(F) - a \otimes I)^*. \tag{3.3}$$

Indeed, on the block-column space the norm is $\|B\|_C^2 = \operatorname{Tr} B^* C^{-1} B$. Its unrestricted dual norm at T is $\operatorname{Tr} CTT^*$. The Hilbert–Schmidt annihilator of the blockwise trace-zero subspace is precisely $\{a \otimes I : a \in \mathbb{C}^k\}$. The norm of the restricted functional is the distance to that annihilator in the dual norm, which proves (3.3). This argument works for complex columns; no Hermiticity assumption on B_i is made.

The unrestricted output dual norm at F is

$$\operatorname{Tr} C' F F^* = \operatorname{Tr} C\, \Phi^{(k)}(F F^*).$$

Dualizing the norm of $B \mapsto B'$ therefore expresses (3.2), for fixed C, as the supremum over nonzero F of the right side of (3.3) divided by this last quantity. Apply (3.1) to the tuple F. Taking its expectation in C proves $\eta_{\mathrm{rect}} \le \eta_{\mathrm{G}}$. Scaling C to trace one does not change either ratio.

Conversely, fix $0 \le e < \eta_{\mathrm{G}}$ and use the original basis tuple, so $k = n$. For a density C on $\mathbb{C}^n \otimes A$ set

$$f(C, a) = \operatorname{Tr} C\big[(\mathbf{A} - aI)(\mathbf{A} - aI)^* - e\mathbf{G}\big].$$

This function is continuous and affine in C, and continuous and convex in the real and imaginary parts of a. The density set is compact and convex; the center space is convex. Sion's minimax theorem [15, Theorem 4.2] applies and gives

$$\sup_C \inf_a f(C, a) = \inf_a \lambda_{\max}\big[(\mathbf{A} - aI)(\mathbf{A} - aI)^* - e\mathbf{G}\big] > 0. \tag{3.4}$$

To justify the strict sign, the expression on the right is bounded below by $(\|a\| - \|\mathbf{A}\|)^2 - e\|\mathbf{G}\|$ when $\|a\|$ is large. It is coercive and attains its minimum. A nonpositive minimum would make (3.1) feasible at e, a contradiction.

Choose C for which $\inf_a f(C, a) > 0$. We can make it faithful. The function $C \mapsto \inf_a f(C, a)$ is concave and is at least $-e\|\mathbf{G}\|$ on every density. Mixing C with a sufficiently small positive multiple of the maximally mixed density thus preserves the strict sign. For the resulting faithful C, (3.3) and norm duality supply a trace-zero block column whose ratio exceeds e. Since e can approach η_{G}, the reverse inequality follows. If $\eta_{\mathrm{G}} = 0$ it is already immediate. This construction uses $k = n = m^2$. □

3.3 Stinespring centering

Choose a minimal Stinespring isometry $V : A \to B \otimes E$ with

$$\Phi(Y) = V^*(Y \otimes I_E)V.$$

In finite dimensions it can be obtained directly from a linearly independent Kraus family, or from the positive Choi matrix. We use these standard representations in the conventions of [16, 3].

Lemma 3.2 (Norm form of the centered Gram problem)**.** *The Gram condition* (3.1) *at e is equivalent to the existence of $K : B \otimes E \to A$ and a complex-linear scalar functional α such that*

$$\|K\|^2 \le e, \qquad K(Y \otimes I_E)V = \Phi(Y) - \alpha(Y)I_A \quad \text{for every } Y. \tag{3.5}$$

If there is a normalized map S satisfying

$$S(Y \otimes I_E)V = \alpha(Y)I_A, \qquad SV = I_A, \tag{3.6}$$

then

$$\eta_{\mathrm{G}} = 1 - \frac{1}{\inf_S \|S\|^2}. \tag{3.7}$$

If no such normalized map exists, $\eta_{\mathrm{G}} = 1$.

Proof. Let $T_i = (F_i^* \otimes I_E)V$ and let $T : \bigoplus_{i=1}^n A \to B \otimes E$ be their row. Then $T^*T = \mathbf{G}$. Put $Q = \mathbf{A} - aI$. The inequality $QQ^* \preceq eT^*T$ says that $K(Tv) = Q^*v$ is well defined and has norm at most $\sqrt{e}$. Extend this map by zero on $(\operatorname{ran} T)^\perp$. Thus $K(F_i^* \otimes I)V = \Phi(F_i^*) - \overline{a_i}I$. The adjoint basis is a complex basis, so its values define α uniquely. Conversely, (3.5) on that basis gives the Gram inequality by taking the squared norm of every row combination. This proves the first assertion without an implicit factorization theorem.

Put $L = V^* - K$. It obeys $L(Y \otimes I)V = \alpha(Y)I$. If $t = \alpha(I) = 0$, then $LV = 0$ and

$$\|V^* - L\|^2 = \|I + LL^*\| = 1 + \|L\|^2 \ge 1.$$

If $t \neq 0$, write $L = tS$, so that $SV = I$, and decompose $S = V^* + S_\perp$, where $S_\perp V = 0$. Orthogonality gives

$$\|V^* - tS\|^2 = |1 - t|^2 + |t|^2\|S_\perp\|^2, \tag{3.8}$$

$$\|S\|^2 = 1 + \|S_\perp\|^2. \tag{3.9}$$

For fixed $r = \|S_\perp\|^2$, the first expression has minimum $r/(1 + r)$ over complex t, attained at $t = 1/(1 + r)$. Minimizing over normalized S proves (3.7). If none exists, all feasible norms below one have been excluded, whereas $e = 1$ was feasible. □

3.4 The complex center is optimally positive

Lemma 3.3 (Inverse Choi and polar reduction)**.** *The infimum over normalized centers in* (3.7) *equals the minimum in* (2.8)*. In particular,*

$$\eta_{\mathrm{G}} = 1 - \frac{1}{M(\mathcal{N})}.$$

Proof. Write $V = \sum_j K_j \otimes |j\rangle$ and $S = \sum_j L_j^* \otimes \langle j|$, with the K_j linearly independent. Use the vectorization $|K\rangle\!\rangle = \sum_i |i\rangle \otimes K|i\rangle$ and let M_K, M_L have these vectors as columns. Expanding the matrix entries of (3.6) shows that, for a possibly complex H,

$$M_L M_K^* = I_A \otimes H, \qquad \operatorname{Tr} H = 1. \tag{3.10}$$

More explicitly, the coefficient of $Y_{\mu\nu}$ in $\sum_j L_j^* Y K_j$ is $\sum_j \overline{(L_j)_{\mu a}}(K_j)_{\nu b}$. Equation (3.10) is precisely the statement that this is $\delta_{ab}\overline{H_{\mu\nu}}$. Thus the associated functional is $\alpha(Y) = \operatorname{Tr}(H^* Y)$.

Since $J = M_K M_K^*$ and M_K has independent columns, (3.10) has a solution exactly when

$$\operatorname{ran}(I_A \otimes H^*) \subseteq \operatorname{ran} J, \tag{3.11}$$

and that solution is $M_L = (I_A \otimes H) J^+ M_K$. Consequently

$$\|S\|^2 = \|\operatorname{Tr}_B[(I \otimes H) J^+ (I \otimes H^*)]\|_\infty. \tag{3.12}$$

The remaining A matrix equals $(SS^*)^{\mathsf{T}}$ under this vectorization, which explains why its operator norm is the stated one.

Partial-trace cyclicity in the B factor rewrites the matrix inside the norm as $\operatorname{Tr}_B[J^+(I \otimes H^* H)]$. Also, $\operatorname{ran} H^* = \operatorname{ran} |H|$. Let $a = \operatorname{Tr}|H| \geq |\operatorname{Tr} H| = 1$. Replacing H by the density $\sigma = |H|/a$ preserves (3.11) and divides the positive matrix in (3.12) by a^2. This is the required positivity argument; restricting a complex center to a Hermitian one without it would not suffice.

Conversely, every admissible density σ supplies a normalized center by (3.10). The admissible densities form a closed subset of the compact density set, and their objective is continuous, because J is fixed. If this subset is nonempty, its minimum is attained. Since $SV = I$, $\|S\| \geq 1$, proving $M \geq 1$.

Finally, for a positive J and Hermitian A_0, the generalized Schur complement gives

$$\begin{pmatrix} J & A_0 \\ A_0 & Z \end{pmatrix} \succeq 0 \quad \Longleftrightarrow \quad \operatorname{ran} A_0 \subseteq \operatorname{ran} J, \quad Z \succeq A_0 J^+ A_0. \tag{3.13}$$

To verify it, test the block form on vectors in $\ker J$ to obtain the range condition. On $\operatorname{supp} J$, complete the square in the first block; the residual form is $Z - A_0 J^+ A_0$. This proves both directions, including singular J. Apply (3.13) to $A_0 = I_A \otimes \sigma$. The partial trace is positive, and equality is attained by $Z = A_0 J^+ A_0$, so the inverse formulation and (2.6) agree. □

Combining lemmas 3.1 to 3.3 yields the finite identity that will be used in both directions of the entropy theorem:

$$\eta_{\mathrm{rect}} = \eta_{\mathrm{G}} = 1 - \beta(\mathcal{N}). \tag{3.14}$$

4 A signed lift and the universal upper bound

This section applies the signed-observable method to a particular comparison channel: erasure. The distinction between observable reconstruction and a physical channel is crucial. The constructed map is linear and adjoint preserving, but need not preserve positive observables.

4.1 Both square inequalities

Assume first $1 < M < \infty$. Choose an optimal positive center σ and the normalized map S from lemma 3.3. Set

$$Q = S - V^*, \qquad QV = 0, \qquad r = \|Q\|^2 = M - 1, \qquad \alpha(Y) = \operatorname{Tr} \sigma Y.$$

The defining equation for S gives

$$\Phi(Y) - \alpha(Y)I = -Q(I - VV^*)(Y \otimes I_E)V.$$

Taking its square and using $Q^*Q \preceq rI$ proves

$$(\Phi(Y) - \alpha(Y)I)^*(\Phi(Y) - \alpha(Y)I) \preceq r\,[\Phi(Y^*Y) - \Phi(Y)^*\Phi(Y)], \tag{4.1}$$

$$(\Phi(Y) - \alpha(Y)I)(\Phi(Y) - \alpha(Y)I)^* \preceq r\,[\Phi(YY^*) - \Phi(Y)\Phi(Y)^*]. \tag{4.2}$$

The second follows by applying the first to Y^*; positivity of σ makes α adjoint preserving. Tensoring Q, V with I_R proves both inequalities for every finite reference and every operator on RB. The amplified center is $(\mathrm{id}_R \otimes \alpha)(Y) \otimes I_A$; it need not commute with the other reference operators.

Put $e = r/(1+r) = 1 - \beta$ and define the erasure channel, with retention probability e, by

$$\mathcal{E}_e(X) = eX \oplus (1 - e) \operatorname{Tr} X.$$

The erasure output algebra is $\mathcal{L}(A) \oplus \mathbb{C}$. Define the observable map $\mathcal{J} : \mathcal{L}(B) \to \mathcal{L}(A) \oplus \mathbb{C}$ by

$$(\mathcal{J}Y)_{\mathrm{r}} = \frac{\Phi(Y) - (1-e)\alpha(Y)I}{e}, \qquad (\mathcal{J}Y)_{\mathrm{e}} = \alpha(Y). \tag{4.3}$$

Then $\mathcal{E}_e^* \mathcal{J} = \Phi$, equivalently $\mathcal{J}^* \mathcal{E}_e = \mathcal{N}$. Direct expansion, valid also for noncommuting amplified centers, gives

$$\begin{aligned} &\Phi(Y^*Y) - \mathcal{E}_e^*((\mathcal{J}Y)^*(\mathcal{J}Y)) \\ &\quad = \Phi(Y^*Y) - \Phi(Y)^*\Phi(Y) - \frac{1-e}{e}(\Phi(Y) - \alpha(Y)I)^*(\Phi(Y) - \alpha(Y)I) \succeq 0. \end{aligned} \tag{4.4}$$

There is a second identity with YY^* and the opposite square order. The final sign in (4.4) is exactly (4.1), since $(1-e)/e = 1/r$.

For a positive matrix ω, write $L_\omega(X) = \omega X$ and $R_\omega(X) = X\omega$. Taking expectations of the two amplified identities in an arbitrary faithful state ω_{RA} yields the separate weighted inequalities

$$\mathcal{J}_R^* L_{\mathcal{E}_{e,R}(\omega)} \mathcal{J}_R \preceq L_{\mathcal{N}_R(\omega)}, \qquad \mathcal{J}_R^* R_{\mathcal{E}_{e,R}(\omega)} \mathcal{J}_R \preceq R_{\mathcal{N}_R(\omega)}. \tag{4.5}$$

The YY^* identity gives the left weight and the Y^*Y identity gives the right weight. Each holds for every ω with the *same* map $\mathcal{J}_R = \mathrm{id}_R \otimes \mathcal{J}$. This state independence will also allow different states in the two weights in section 7.

4.2 Inverse quadratic duality and the entropy metric

For completeness, we prove the exact comparison mechanism rather than treating the signed map as a simulator. This is the mechanism of [17, Lemma S6 and Theorem S5], in the erasure specialization.

Lemma 4.1 (Inverse quadratic comparison)**.** *Let A, B be positive definite operators on finite Hilbert spaces and T a linear map with $T^*AT \preceq B$. Then*

$$\langle T^*z, B^{-1}T^*z\rangle \le \langle z, A^{-1}z\rangle.$$

Proof. Completing the square gives

$$\langle z, A^{-1}z\rangle = \sup_x \{2\,\mathrm{Re}\langle x, z\rangle - \langle x, Ax\rangle\}.$$

Restrict this supremum to $x = Ty$ and use $\langle Ty, ATy\rangle \le \langle y, By\rangle$. The resulting supremum over y equals $\langle T^*z, B^{-1}T^*z\rangle$ by the same identity. □

For $\omega \succ 0$ and Hermitian H, define

$$g_\omega(H) = \int_0^1 \langle H, [(1-u)L_\omega + uR_\omega]^{-1}H\rangle\,\mathrm{d}u. \tag{4.6}$$

In an eigenbasis of ω, with eigenvalues $\lambda_i > 0$,

$$g_\omega(H) = \sum_{i,j} |H_{ij}|^2 \frac{\log\lambda_i - \log\lambda_j}{\lambda_i - \lambda_j}, \tag{4.7}$$

where the divided difference is $1/\lambda_i$ on the diagonal. Indeed, integrating $[(1-u)\lambda_i + u\lambda_j]^{-1}$ proves this formula. It is the Hessian of $F(\omega) = \mathrm{Tr}\,\omega\log\omega$, as follows by differentiating the spectral logarithm. For example, the derivative formula

$$D\log_\omega(H) = \int_0^\infty (\omega + sI)^{-1}H(\omega + sI)^{-1}\,\mathrm{d}s$$

has exactly the divided differences in (4.7). Thus g_ω is the Bogoliubov–Kubo–Mori (BKM) metric.

Add the inequalities (4.5) with weights $1-u, u$ and apply lemma 4.1 to $z = \mathcal{E}_{e,R}(H)$. The reconstruction identity gives $\mathcal{J}_R^* z = \mathcal{N}_R(H)$. Integrating in u proves

$$g_{\mathcal{N}_R(\omega)}(\mathcal{N}_R(H)) \le g_{\mathcal{E}_{e,R}(\omega)}(\mathcal{E}_{e,R}(H)). \tag{4.8}$$

All inverses here act on the actual output supports; the two erasure branches are faithful when $0 < e < 1$.

Lemma 4.2 (Affine-path integration and finite-entropy boundaries)**.** *For faithful states ρ, τ, put $H = \rho - \tau$. Then*

$$D(\rho\|\tau) = \int_0^1 (1-s)\, g_{\tau+sH}(H)\,\mathrm{d}s. \tag{4.9}$$

If $D(\rho\|\tau) < \infty$ and ζ is faithful, the common mixtures $\rho_\epsilon = (1-\epsilon)\rho + \epsilon\zeta$ and $\tau_\epsilon = (1-\epsilon)\tau + \epsilon\zeta$ satisfy $D(\rho_\epsilon\|\tau_\epsilon) \to D(\rho\|\tau)$. The same convergence holds after any CPTP map, on its actual output support.

Proof. Relative entropy is the Bregman remainder of F on the trace-one affine space. Its first derivative remainder vanishes, and Taylor's integral formula gives (4.9).

For the boundary assertion, finite relative entropy implies $\rho \preceq c\tau$ for some finite c. Hence $\rho_\epsilon \preceq \max(c,1)\tau_\epsilon$. The term $\mathrm{Tr}\,\rho_\epsilon\log\rho_\epsilon$ converges because $x\log x$ is continuous at zero. Diagonalize τ_ϵ. For any eigenvalue $\lambda_i(\epsilon) \to 0$, its diagonal weight $w_i(\epsilon)$ in ρ_ϵ is at most $\max(c,1)\lambda_i(\epsilon)$, so $w_i(\epsilon)|\log\lambda_i(\epsilon)| \to 0$. On the spectral subspace of eigenvalues bounded away from zero, continuous functional calculus gives convergence to $\mathrm{Tr}\,\rho\log\tau$. There are finitely many eigenvalues, so the cross term converges as well. A channel preserves $\rho \preceq c\tau$; its common mixture state is faithful on the fixed output support. The same argument applies there. □

4.3 The universal Umegaki inequality

Integrate (4.8) along the path in (4.9). The orthogonal erasure branches give

$$D(\mathcal{E}_{e,R}\rho\|\mathcal{E}_{e,R}\tau) = eD(\rho\|\tau) + (1-e)D(\rho_R\|\tau_R).$$

This proves (2.9) for faithful states. Lemma 4.2 extends it to every finite-input-entropy pair; the reference terms converge by the same lemma.

If $M = 1$, (4.1) has $r = 0$, so $\Phi(Y) = \alpha(Y)I$ for every Y. This is a replacer channel $\mathcal{N}(X) = \mathrm{Tr}(X)\sigma$, and (2.9) is equality with $e = 0$. If $M = \infty$, use the identity comparison channel instead of erasure and take $\mathcal{J} = \Phi$. Stinespring Schwarz gives both weighted inequalities (4.5). The preceding metric and path argument proves ordinary data processing, which is (2.9) with $e = 1$. No assertion about a finite inverse-Choi center is needed in this case.

Subtracting $D(\rho_R\|\tau_R)$ now gives $\eta_D^{\Delta} \le 1-\beta$, and equal-reference pairs give $\eta_D^{\mathrm{c}} \le 1-\beta$. For any ρ_{URA}, apply (2.9) with reference UR to the pair

$$\rho_{URA}, \qquad \rho_U \otimes \rho_{RA}.$$

Its input relative entropy is $I(U;RA)$ and its reference relative entropy is $I(U;R)$. Their difference is $I(U;A|R)$, and the output difference is $I(U;B|R)$. Both terms are finite in finite dimensions because the first state is supported in the product of its marginals' supports. This proves $\eta_{\mathrm{condMI}} \le 1-\beta$ for arbitrary quantum U, R, under the explicit definition (2.3).

5 The coherent converse with exactly fixed marginals

The rectangular problem in lemma 3.1 is not itself an entropy ratio. Its conversion into one is the main converse construction. A new reference direction makes a rectangular matrix into an off-diagonal Hermitian perturbation. Making that direction rare selects the inverse quadratic weight needed for the exact optimum.

Lemma 5.1 (A positive coherent family)**.** *Let $C \succ 0$ be a trace-one state on R_0A, let $B : A \to R_0A$ be nonzero with $\mathrm{Tr}_A B = 0$, and let $\tau \succ 0$ be a state on A. Put $R = R_0 \oplus \mathbb{C}r$, $q = 1-p$, and*

$$\omega_p = \begin{pmatrix} qC & 0 \\ 0 & p\tau \end{pmatrix}, \qquad H_p = \begin{pmatrix} 0 & \sqrt{pq}\,B \\ \sqrt{pq}\,B^* & 0 \end{pmatrix}, \qquad \rho_{p,h} = \omega_p + hH_p. \tag{5.1}$$

For $0 < p < 1$, these are faithful states whenever $|h|$ is sufficiently small in an interval independent of p. Both their R and A marginals equal those of ω_p exactly.

Proof. The Schur complement of qC in $\rho_{p,h}$ is

$$p(\tau - h^2 B^* C^{-1} B).$$

It is positive definite, for example, when $h^2\|\tau^{-1/2}B^*C^{-1}B\tau^{-1/2}\| < 1$. This condition is independent of p. The perturbation has total trace zero. Its reference marginal is zero because each block of B has trace zero. Its input marginal is zero because it is off diagonal in the reference direct sum. Thus $\mathrm{Tr}_A H_p = \mathrm{Tr}_R H_p = 0$. □

Lemma 5.2 (The inverse weight selected by a rare sector)**.** *For the family (5.1),*

$$\lim_{p\downarrow 0} \frac{g_{\omega_p}(H_p)}{2p\log(1/p)} = q_C(B). \tag{5.2}$$

The corresponding output limit is $q_{C'}(B')$, where $C' = \mathcal{N}_{R_0}(C)$ and $B' = \mathcal{N}(B)$ blockwise.

Proof. Let $c_i > 0$ and $v_j > 0$ be eigenvalues of C and τ, and write B_{ij} in these eigenbases. Only the cross-sector entries contribute to (4.7); both orientations contribute equally. Hence

$$g_{\omega_p}(H_p) = 2pq \sum_{i,j} |B_{ij}|^2 \frac{\log(qc_i) - \log(pv_j)}{qc_i - pv_j}. \tag{5.3}$$

Coincident arguments are interpreted by continuity. For each fixed (i, j), division by $2p \log(1/p)$ makes its factor converge to $|B_{ij}|^2/c_i$. There are finitely many terms, and their sum is $\operatorname{Tr} B^* C^{-1} B$. The output has the same block form, with faithful C' and $\tau' = \mathcal{N}(\tau)$ on the fixed output support. The identical divided-difference computation proves the second limit. □

Completion of theorem 2.3. For each fixed p, analyticity at the faithful state ω_p gives

$$D(\omega_p + hH_p \| \omega_p) = \frac{h^2}{2} g_{\omega_p}(H_p) + O_p(h^3).$$

The same expansion applies to the output. The denominator's quadratic coefficient is strictly positive. Thus lemma 5.2 gives

$$\lim_{p \downarrow 0} \lim_{h \to 0} \frac{D(\mathcal{N}_R(\omega_p + hH_p) \| \mathcal{N}_R \omega_p)}{D(\omega_p + hH_p \| \omega_p)} = \frac{q_{C'}(B')}{q_C(B)}. \tag{5.4}$$

Every pair on the left has exactly the same reference marginal. Taking the supremum over C, B and using (3.14) proves $\eta_D^{\mathrm{c}} \geq 1 - \beta$.

For clarity, the limit statement produces ordinary finite admissible pairs: given a desired accuracy, first choose C, B with nearly optimal rectangular ratio, then choose a sufficiently small p, and finally a sufficiently small nonzero h. No uniform bound on the $O_p(h^3)$ remainder as $p \downarrow 0$ is claimed or required.

To obtain a conditional-MI witness, let a uniform classical bit U select $\omega_p + hH_p$ and $\omega_p - hH_p$. Its average is exactly ω_p. By lemma 5.1, $I(U; R) = I(U; A) = 0$ exactly. Also,

$$\begin{aligned} I(U; RA) &= \tfrac{1}{2} D(\omega_p + hH_p \| \omega_p) + \tfrac{1}{2} D(\omega_p - hH_p \| \omega_p), \\ I(U; RB) &= \tfrac{1}{2} D(\mathcal{N}_R(\omega_p + hH_p) \| \mathcal{N}_R \omega_p) + \tfrac{1}{2} D(\mathcal{N}_R(\omega_p - hH_p) \| \mathcal{N}_R \omega_p). \end{aligned}$$

Subtracting $I(U; R) = 0$ makes these the two conditional mutual informations. Their ordered local limits give the same rectangular ratio. Thus $\eta_{\mathrm{condMI}} \geq 1 - \beta$. Equal-reference pairs are admissible in the conditional coefficient, so $\eta_D^{\Delta} \geq \eta_D^{\mathrm{c}}$. The upper bounds already proved give all equalities.

Lemma 3.1 supplies $\dim R_0 \leq m^2$; only the one additional direction was used. The stated reference bounds follow. When $d = 1$, every channel is a replacer and every admissible coefficient is zero. For a nontrivial replacer with $d > 1$, admissible finite pairs exist but their output numerator is zero; the same conclusion holds. □

Remark 5.3 (Information in correlations and its size). The binary label in the construction is invisible in each subsystem separately, although it is visible in their correlations. This explains why classical-reference intuition need not give the coefficient. It does not ensure a large absolute signal. For fixed p, the information vanishes quadratically as $h \to 0$, and its quadratic scale is itself $p \log(1/p)$ as the sector becomes rare.

6 Support, harmonic duality and exact products

The inverse-Choi formula has consequences that are easier to see in matrix geometry than in entropy ratios. We develop them explicitly, retaining the positive CP-replacer coefficient as a distinct, previously established object.

6.1 Complete output rows and strict contraction

Let P be the support projection of J and define

$$E_J = \ker \mathrm{Tr}_A(I_{AB} - P). \tag{6.1}$$

This is an output subspace, not an environment or an erasure flag. For $b \in B$,

$$\langle b, \mathrm{Tr}_A(I-P)b\rangle = \sum_{i=1}^{d} \|(I-P)(|i\rangle \otimes b)\|^2.$$

Consequently E_J is the largest subspace with $A \otimes E_J \subseteq \operatorname{ran} J$. A density σ is feasible in (2.8) exactly when its range lies in E_J.

Corollary 6.1 (Strict-contraction criterion)**.** *For every channel, the following are equivalent:*

$$\eta_D^{\mathrm{c}}(\mathcal{N}) < 1, \qquad E_J \neq \{0\}, \qquad J \succeq aI_A \otimes \sigma \text{ for some } \sigma \in \mathsf{D}(B),\ a > 0.$$

The same criterion holds for η_D^{Δ} and η_{condMI}.

Proof. The first two conditions are equivalent by the nonempty feasible set in (2.8). The last implies the range inclusion and hence the second. Conversely, take a unit vector $b \in E_J$ and let $a > 0$ be at most the smallest positive eigenvalue of J. Then $J \succeq aP \succeq aI_A \otimes |b\rangle\langle b|$. □

In channel language the last condition is positive CP-replacer minorization. With the notation of [6, Definition 7, equation (3.25)], put

$$\alpha_+(\mathcal{N}) = \max\{\mathrm{Tr}\, H : H \succeq 0,\ I_A \otimes H \preceq J\}. \tag{6.2}$$

If $\alpha_+ > 0$, writing $H = \alpha_+\sigma$ gives $\mathcal{N} = \alpha_+\mathcal{R}_\sigma + (1-\alpha_+)\mathcal{Q}$ with $\mathcal{Q}$ CPTP, except that at $\alpha_+ = 1$ the channel is itself a replacer. Here $\mathcal{R}_\sigma(X) = \mathrm{Tr}(X)\sigma$. Positivity of H is required. The Doeblin optimization over arbitrary Hermitian H in [6] is a different coefficient.

6.2 The harmonic dual

Assume $E_J \neq 0$. Compress J^+ to the positive definite operator K on $A \otimes E_J$. The compression is positive definite because $A \otimes E_J \subseteq \operatorname{ran} J$; invariance of this subspace under J is not required.

Theorem 6.2 (Matrix harmonic dual)**.** *With $H_\theta = \mathrm{Tr}_A[(\theta \otimes I_{E_J})K]$,*

$$\boxed{\beta(\mathcal{N}) = \min_{\theta \in \mathsf{D}(A)} \mathrm{Tr}\, H_\theta^{-1}.} \tag{6.3}$$

Every H_θ is positive definite, including for singular θ.

Proof. The inverse-Choi objective can be written

$$M = \min_{\sigma \in \mathsf{D}(E_J)} \max_{\theta \in \mathsf{D}(A)} \mathrm{Tr}\, K(\theta \otimes \sigma^2).$$

The function is affine in θ and convex in σ: for fixed θ it is $\mathrm{Tr}\, H_\theta \sigma^2$, a positive quadratic form. Both density sets are compact and convex, so convex minimax applies. If $K \succeq cI$, then $H_\theta \succeq cI$ because

$\operatorname{Tr}\theta = 1$. Here partial-trace cyclicity identifies H_θ with the positive matrix $\operatorname{Tr}_A[(\theta^{1/2}\otimes I)K(\theta^{1/2}\otimes I)]$. Weighted Hilbert–Schmidt Cauchy–Schwarz gives

$$1 = (\operatorname{Tr}\sigma)^2 \le (\operatorname{Tr} H_\theta^{-1})(\operatorname{Tr} H_\theta \sigma^2).$$

Equality holds at $\sigma = H_\theta^{-1}/\operatorname{Tr} H_\theta^{-1}$. Thus

$$M = \max_\theta \frac{1}{\operatorname{Tr} H_\theta^{-1}} = \frac{1}{\min_\theta \operatorname{Tr} H_\theta^{-1}},$$

which proves the formula. □

Theorem 6.3 (Exact tensor products)**.** *For arbitrary finite channels,*

$$\beta(\mathcal{N}_1 \otimes \mathcal{N}_2) = \beta(\mathcal{N}_1)\beta(\mathcal{N}_2), \tag{6.4}$$
$$\eta(\mathcal{N}_1 \otimes \mathcal{N}_2) = 1 - (1-\eta(\mathcal{N}_1))(1-\eta(\mathcal{N}_2)), \tag{6.5}$$

where η is any coefficient in (2.7).

Proof. First the feasible support spaces multiply, including zero spaces. Let $P_i = \operatorname{supp} J_i$, $S_i = \operatorname{Tr}_{A_i} P_i$ and $d_i = \dim A_i$. Since $0 \preceq P_i \preceq I$, $0 \preceq S_i \preceq d_i I$. The space E_{J_i} is the eigenspace of S_i at d_i. After the natural tensor-factor reordering, the product support projection is $P_1 \otimes P_2$ and its output partial trace is $S_1 \otimes S_2$. An eigenvalue product equals $d_1 d_2$ exactly when each factor equals d_i. Hence

$$E_{J_1 \otimes J_2} = E_{J_1} \otimes E_{J_2}.$$

If a factor is zero, the product β is zero and (6.4) follows.

Otherwise, product choices of σ in the primal give $M_{12} \le M_1 M_2$, since their inverse forms and positive partial-trace matrices tensor. Therefore $\beta_{12} \ge \beta_1\beta_2$. On the product support the compressed inverse is $K_1 \otimes K_2$. Choose minimizing θ_1, θ_2 in (6.3). Their product produces $H_{\theta_1} \otimes H_{\theta_2}$, so the dual gives $\beta_{12} \le (\operatorname{Tr} H_{\theta_1}^{-1})(\operatorname{Tr} H_{\theta_2}^{-1}) = \beta_1\beta_2$. The entropy identity follows from theorem 2.3. □

This is an all-pairs, variable-reference product law. It does not assert tensorization of an ordinary coefficient or a fixed-equilibrium entropy-decay constant; the latter belong to a different setting [5].

6.3 A sharp comparison with positive replacer weight

Theorem 6.4 (Harmonic bounds and sharp Doeblin comparison)**.** *If $E_J \neq 0$, set*

$$K_E = P_{E_J}(\operatorname{Tr}_A J^+)P_{E_J}\Big|_{E_J}, \qquad h(J) = \operatorname{Tr}_{E_J} K_E^{-1}.$$

If $E_J = 0$, put $h(J) = 0$. Then

$$h(J) \le \beta(\mathcal{N}) \le d\, h(J), \qquad \boxed{\alpha_+(\mathcal{N}) \le \beta(\mathcal{N}) \le d^2 \alpha_+(\mathcal{N}).} \tag{6.6}$$

The factor d^2 cannot be decreased uniformly in channels of input dimension $d \ge 2$.

Proof. The zero-space case follows from corollary 6.1. For feasible σ write $T_\sigma = \operatorname{Tr}_B[(I\otimes\sigma)J^+(I\otimes\sigma)]$. It is positive and $\operatorname{Tr} T_\sigma = \operatorname{Tr} K_E\sigma^2$. Its norm lies between its trace divided by d and its trace. The minimum possible trace is $1/h(J)$, attained at $\sigma = K_E^{-1}/\operatorname{Tr} K_E^{-1}$ by Cauchy–Schwarz. Thus $1/(dh) \le M \le 1/h$, which proves the first bounds.

If $J \succeq aI \otimes \sigma$, $a > 0$, then

$$\begin{pmatrix} J & I \otimes \sigma \\ I \otimes \sigma & (I \otimes \sigma)/a \end{pmatrix} \succeq 0.$$

Indeed, subtract $\operatorname{diag}(J - aI \otimes \sigma, 0)$; the remaining matrix is $I \otimes \sigma$ tensored with the positive scalar matrix $\left(\begin{smallmatrix} a & 1 \\ 1 & 1/a \end{smallmatrix}\right)$. This is feasible in (2.6) with $t = 1/a$, proving $\alpha_+ \leq \beta$.

For a positive operator T on AB one has

$$T \preceq dI_A \otimes \operatorname{Tr}_A T. \tag{6.7}$$

For a rank-one term $|v\rangle\langle v|$, conjugation by the inverse square root of its B marginal on that marginal's support makes the vector's squared norm its Schmidt rank, at most d. This proves (6.7) for that term. Sum rank-one terms to obtain the stated inequality.

When $J \succ 0$, apply (6.7) to J^{-1} and invert. It gives $J \succeq d^{-1} I_A \otimes (\operatorname{Tr}_A J^{-1})^{-1}$, so $\alpha_+ \geq h/d$. For singular J, let $J_\epsilon = J + \epsilon I$. Then

$$\operatorname{Tr}_A J_\epsilon^{-1} = \epsilon^{-1} \operatorname{Tr}_A (I - P) + \operatorname{Tr}_A J^+ + O(\epsilon).$$

Decompose $B = E_J \oplus E_J^\perp$. The leading matrix is zero on the first block and positive definite on the second. The block inverse, or its Schur complement, consequently converges to $P_{E_J} K_E^{-1} P_{E_J}$: its E_J block converges to K_E^{-1}, and the other blocks vanish. Taking the limit in the preceding feasible matrix inequality again gives $\alpha_+ \geq h/d$. Combine this with $\beta \leq dh$ to prove $\beta \leq d^2 \alpha_+$.

For sharpness, use any orthogonal unitary error basis with probabilities $p_g > 0$. The calculation in theorem 10.1 gives $\beta = d^4 / \sum_g p_g^{-1}$. Also, $\alpha_+ = d^2 \min_g p_g$: testing $I \otimes H \preceq J$ against each maximally entangled error-basis vector gives $\operatorname{Tr} H/d \leq d p_g$, and scalar H attains the minimum bound. Let one probability tend to zero while the others stay bounded away from zero. Then $\beta/\alpha_+ \to d^2$. □

The qualitative minorization criterion and its information-theoretic applications are prior work [6, 9]. The comparison (6.6) quantifies how the exact coefficient can improve that baseline. Its factor is dimension dependent; for n qubits it is 4^n.

6.4 Prescribed input marginals

Theorem 6.5 (Faithful marginal padding and support dependence)**.** *Let ω_A be faithful. The value of $\eta_D^c(\mathcal{N})$ is unchanged if both compared input marginals are required to equal ω_A. Reference dimension at most $m^2 + 2$ suffices to approach it. The binary witnesses for η_{condMI} may also have $\rho_A = \omega_A$ and $I(U;A) = I(U;R) = 0$ exactly.*

For nonfaithful ω_A, either fixing both input marginals to ω_A, or fixing only the second input marginal in (2.1), gives the coefficient of the restricted channel $\mathcal{N}|_{\mathcal{L}(\operatorname{supp}\omega_A)}$. Thus these values depend on ω_A only through its support.

Proof. The witnesses in section 5 already have a common input marginal, say ν_A. Choose $0 < a < 1$ so small that $\tau_A = (\omega_A - a\nu_A)/(1-a) \succeq 0$. Append one orthogonal reference flag and put

$$\widetilde{\rho} = a\rho \oplus (1-a)\tau_A, \qquad \widetilde{\sigma} = a\sigma \oplus (1-a)\tau_A.$$

The reference marginals agree, and both input marginals are exactly ω_A. Orthogonal direct sums give $D(\widetilde{\rho}\|\widetilde{\sigma}) = aD(\rho\|\sigma)$ before and after the channel, so the ratio is unchanged. The same padding, with the same weights in both bit branches, preserves the two independence identities and scales both Holevo informations by a. This uses one additional reference direction.

If $S = \operatorname{supp}\omega_A$, positivity forces every joint state with that input marginal to be supported in $R \otimes S$. If only the second input marginal is fixed, finite $D(\rho\|\sigma)$ forces the first joint state into that support as well. The upper bound is therefore the restricted channel's coefficient. On S, ω_A is faithful, so the preceding padding construction gives the matching lower bound. A one-dimensional support gives zero. □

The comparison state in this theorem can be correlated with its reference. It is not fixed to $\rho_R \otimes \omega_A$. There is consequently no conflict with the distinct product-reference coefficient in table 2, or with entropy contraction toward a specified equilibrium. In particular, fixing only the second input marginal is the input-dependent complete coefficient of [6, Definition 12]; for faithful ω_A it too has the common value in (2.7).

7 Universality for regular modular Petz divergences

The entropy upper bound used a single baseline state in its two weighted inequalities. Their state independence permits a stronger use: the left weight can come from one state and the right weight from another. This controls resolvent divergences directly, without first integrating an entropy Hessian.

7.1 The divergence and its exact generator class

Definition 7.1 (Regular modular Petz generators)**.** Let $f : (0,\infty) \to \mathbb{R}$ be nonaffine and operator convex, with

$$f(1) = f'(1) = 0, \qquad f(0+) < \infty, \qquad \ell_f = \lim_{x\to\infty} \frac{f(x)}{x} < \infty.$$

Its modular Petz divergence on faithful states is

$$D_f(\rho\|\sigma) = \langle \sigma^{1/2}, f(L_\rho R_\sigma^{-1})\sigma^{1/2}\rangle. \tag{7.1}$$

For positive semidefinite states use its continuous spectral extension: if $\rho = \sum_i r_i |u_i\rangle\langle u_i|$ and $\sigma = \sum_j s_j |v_j\rangle\langle v_j|$, then

$$D_f(\rho\|\sigma) = \sum_{i,j} F_f(r_i, s_j)|\langle u_i, v_j\rangle|^2, \qquad F_f(a,b) = bf(a/b) \quad (a,b>0),$$

with $F_f(0,b) = bf(0+)$, $F_f(a,0) = a\ell_f$ and $F_f(0,0) = 0$.

This is the modular, or standard Petz, divergence [14]. It is not the maximal quantum f-divergence and not the hockey-stick integral construction of [10]. In particular, an equality for another divergence with the same classical restriction would not imply the theorem below.

For any divergence V used here, define η_V^{c} and η_V^{Δ} by replacing D in (2.1) and (2.2). Define, explicitly,

$$\mathcal{I}_V(U;A|R)_\rho = V(\rho_{URA}\|\rho_U \otimes \rho_{RA}) - V(\rho_{UR}\|\rho_U \otimes \rho_R), \tag{7.2}$$

and let $\eta_{\mathcal{I}_V}$ be the output-to-input ratio of this quantity, with a strictly positive finite input and finite terms. All auxiliary systems are arbitrary finite quantum systems. For $V = D$ this is ordinary conditional mutual information. For other V it denotes exactly the displayed difference, without an optimization over comparison states.

Theorem 7.2 (Exact regular modular Petz coefficients)**.** *For every generator in definition 7.1 and every finite channel,*

$$\eta^{\mathrm{c}}_{D_f}(\mathcal{N}) = \eta^{\Delta}_{D_f}(\mathcal{N}) = \eta_{\mathcal{I}_{D_f}}(\mathcal{N}) = 1 - \beta(\mathcal{N}). \tag{7.3}$$

For arbitrary state pairs and references,

$$D_f(\mathcal{N}_R\rho\|\mathcal{N}_R\sigma) \le (1-\beta)D_f(\rho\|\sigma) + \beta D_f(\rho_R\|\sigma_R). \tag{7.4}$$

The sharpness bounds $\dim R \le m^2 + 1$ *and a binary classical auxiliary independent of both local marginals remain valid.*

7.2 Positive resolvents and endpoint integrability

Lemma 7.3 (The precise integral representation)**.** *For* f *in definition 7.1 there is a positive measure* ν *on* $(0,\infty)$ *such that*

$$f(x) = \int_{(0,\infty)} \frac{(x-1)^2}{x+s}\,\mathrm{d}\nu(s), \qquad \int_{(0,\infty)} (1+s^{-1})\,\mathrm{d}\nu(s) = f(0+) + \ell_f =: k_f \in (0,\infty). \tag{7.5}$$

Proof. Franz–Hiai's representation [4, Theorem 2.2], at the base point one, is

$$f(x) = \gamma(x-1)^2 + \int_{[0,\infty)} \frac{(x-1)^2}{(x+s)(1+s)^2}\,\mathrm{d}\mu(s), \qquad \gamma \ge 0, \quad \int (1+s)^{-3}\,\mathrm{d}\mu(s) < \infty.$$

The affine part is zero by normalization. Finite recession slope forces $\gamma = 0$; finite $f(0+)$ excludes an atom at zero. Set $\mathrm{d}\nu(s) = (1+s)^{-2}\,\mathrm{d}\mu(s)$. As $x \downarrow 0$, the nonnegative integrands increase to $1/s$. As $x \uparrow \infty$, their ratios to x increase to one for $x \ge 1$. Monotone convergence gives $f(0+) = \int s^{-1}\,\mathrm{d}\nu$ and $\ell_f = \int \mathrm{d}\nu$. Nonaffineness means the measure is nonzero, proving strict positivity. □

Let $f_s(x) = (x-1)^2/(x+s)$. Commutation of left and right multiplication in the modular calculus gives

$$D_{f_s}(\rho\|\sigma) = Q_s(\rho,\sigma) := \langle \rho - \sigma, (L_\rho + sR_\sigma)^{-1}(\rho - \sigma)\rangle \quad (s > 0). \tag{7.6}$$

To check it, move $\Delta - I$, with $\Delta = L_\rho R_\sigma^{-1}$, onto $\sigma^{1/2}$ in (7.1); its value is $(\rho - \sigma)\sigma^{-1/2}$. The factors of $R_\sigma^{1/2}$ cancel in $(\Delta + sI)^{-1} = R_\sigma(L_\rho + sR_\sigma)^{-1}$, giving (7.6). Thus $D_f = \int Q_s\,\mathrm{d}\nu(s)$ for faithful states.

7.3 Two-state weights give the upper inequality

Write $e = 1 - \beta$ and initially suppose $0 < e < 1$. In (4.5), use ρ for the left-weight state and σ for the right-weight state. Adding with weight $s > 0$ gives

$$\mathcal{J}_R^*(L_{\mathcal{E}_{e,R}(\rho)} + sR_{\mathcal{E}_{e,R}(\sigma)})\mathcal{J}_R \preceq L_{\mathcal{N}_R(\rho)} + sR_{\mathcal{N}_R(\sigma)}.$$

Apply lemma 4.1 with $z = \mathcal{E}_{e,R}(\rho - \sigma)$ and use reconstruction. This proves

$$Q_s(\mathcal{N}_R\rho, \mathcal{N}_R\sigma) \le Q_s(\mathcal{E}_{e,R}\rho, \mathcal{E}_{e,R}\sigma).$$

The two erasure branches are orthogonal, and Q_s is homogeneous of degree one in the pair. Consequently the right side is $eQ_s(\rho,\sigma) + (1-e)Q_s(\rho_R,\sigma_R)$. Positive integration proves (7.4). For

$e = 1$ take the identity comparison and $\mathcal{J} = \Phi$, as in section 4; for $e = 0$ the replacer gives equality with the reference divergence.

These statements extend continuously to all PSD state pairs. For the scalar perspective of f_s,

$$0 \le \frac{(a-b)^2}{a+sb} \le a + \frac{b}{s} \quad (a, b \ge 0), \tag{7.7}$$

with $0/0 = 0$. It is continuous, including at $(0, 0)$. Spectral calculus for the commuting operators L_ρ, R_σ therefore gives continuity of Q_s. Equivalently one can approximate this continuous scalar function uniformly by polynomials on the compact spectral square. Its spectral expression and (7.7) bound Q_s by $1 + s^{-1}$. Lemma 7.3 supplies an integrable bound, so dominated convergence proves continuity of D_f and the boundary passage in (7.4).

Subtracting the reference term proves the two relative-divergence upper bounds. Applying the same inequality with reference UR to $(\rho_{URA}, \rho_U \otimes \rho_{RA})$ proves the upper bound for (7.2). The reference subtraction is legitimate even when U is quantum.

7.4 The matching local and coherent converse

Lemma 7.4 (Regular Petz coherent Hessians)**.** *For (5.1) at each fixed p,*

$$D_f(\omega_p + hH_p \| \omega_p) = \frac{h^2}{2} g^f_{\omega_p}(H_p) + o_p(h^2),$$

and

$$\lim_{p \downarrow 0} \frac{g^f_{\omega_p}(H_p)}{2p} = k_f q_C(B). \tag{7.8}$$

The output limit is $k_f q_{C'}(B')$.

Proof. For one resolvent, (7.6) gives exactly

$$Q_s(\omega + hH, \omega) = h^2 \langle H, (L_{\omega+hH} + sR_\omega)^{-1} H \rangle.$$

At fixed faithful ω, choose h so that $\omega + hH \succeq \omega/2$. The right quadratic form is then at most twice its value with $\omega + hH$ replaced by ω. This bounds the second-order quotient by an integrable function of s. Taking $h \to 0$ under the positive integral is therefore justified. For the coherent block family, diagonalization as in lemma 5.2 gives

$$g^{f_s}_{\omega_p}(H_p) = 2pq \sum_{i,j} |B_{ij}|^2 \left(\frac{1}{qc_i + spv_j} + \frac{1}{pv_j + sqc_i} \right). \tag{7.9}$$

After division by $2p$, its summand is bounded above by $(1 + s^{-1})|B_{ij}|^2/c_i$ and converges to that same quantity as $p \downarrow 0$. There are finitely many (i, j). Dominated convergence against ν proves (7.8). Faithfulness of C', τ' on the actual output support gives the identical proof at the output. □

Completion of theorem 7.2. Take $h \to 0$ at each fixed p, and then $p \downarrow 0$. The common positive factor k_f in (7.8) cancels, leaving $q_{C'}(B')/q_C(B)$. The same-reference lower bound is therefore $\eta_{\text{rect}} = 1 - \beta$.

For the binary auxiliary of section 5, the compared joint states are the two equally weighted blocks and their common average. The direct-sum formula for D_f makes $\mathcal{I}_{D_f}(U; A|R)$ the average of the two local divergences; the reference term vanishes exactly. Both signs have the same second-order coefficient in lemma 7.4. This gives the matching information lower bound. The conditional

coefficient contains the equal-reference pairs. Together with the upper bounds already proved, this establishes (7.3) with the stated reference dimensions. Empty one-dimensional cases have value zero. □

Umegaki entropy is not a member of definition 7.1: the normalized generator $x \log x - x + 1$ has infinite recession slope. Its independent proof in Sections 4–5 is therefore necessary.

8 Petz Rényi and quantum Chernoff contraction

Taking a logarithm is not innocuous for contraction coefficients. We show the required scalar inequality and then prove sharpness locally. For Chernoff divergence, an additional compact-order Taylor estimate justifies the optimization over the order.

For $0 < \alpha < 1$, define

$$Q_\alpha(\rho, \sigma) = \operatorname{Tr} \rho^\alpha \sigma^{1-\alpha}, \qquad D_\alpha(\rho\|\sigma) = -\frac{1}{1-\alpha} \log Q_\alpha(\rho, \sigma).$$

Define quantum Chernoff divergence by

$$C_Q(\rho, \sigma) = -\log \inf_{0<\alpha<1} Q_\alpha(\rho, \sigma).$$

If the logarithm's argument is zero the value is $+\infty$. The complete and reference-subtracted coefficients, and the generalized information coefficient, use precisely (7.2) and the finiteness conventions in section 2. In particular, $\mathcal{I}_{D_\alpha}$ is not an optimized or sandwiched Rényi conditional mutual information.

Theorem 8.1 (Logarithmic extensions)**.** *For every fixed $0 < \alpha < 1$ and every finite channel,*

$$\eta^{\mathrm{c}}_{D_\alpha}(\mathcal{N}) = \eta^{\Delta}_{D_\alpha}(\mathcal{N}) = \eta_{\mathcal{I}_{D_\alpha}}(\mathcal{N}) = 1 - \beta(\mathcal{N}), \tag{8.1}$$

$$\eta^{\mathrm{c}}_{C_Q}(\mathcal{N}) = \eta^{\Delta}_{C_Q}(\mathcal{N}) = \eta_{\mathcal{I}_{C_Q}}(\mathcal{N}) = 1 - \beta(\mathcal{N}). \tag{8.2}$$

Both divergences V satisfy, on their finite-input-divergence domains,

$$V(\mathcal{N}_R\rho\|\mathcal{N}_R\sigma) \le (1-\beta)V(\rho\|\sigma) + \beta V(\rho_R\|\sigma_R). \tag{8.3}$$

A reference of dimension at most $m^2 + 1$ and a binary classical auxiliary independent of each local marginal suffice for sharpness.

8.1 The logarithmic upper bound

For fixed α, the normalized generator

$$f_\alpha(x) = \frac{1 - \alpha + \alpha x - x^\alpha}{1 - \alpha}$$

belongs to definition 7.1. Operator convexity follows from operator concavity of x^α; also $f_\alpha(0+) = 1$ and $\ell_{f_\alpha} = \alpha/(1-\alpha)$. Its modular divergence is $D_{f_\alpha} = (1 - Q_\alpha)/(1-\alpha)$. For clarity, the operator concavity of the power follows from the convergent integral

$$x^\alpha = \frac{\sin(\pi\alpha)}{\pi} \int_0^\infty \frac{x}{x+s} s^{\alpha-1}\, \mathrm{d}s.$$

Each matrix function $X(X+sI)^{-1} = I - s(X+sI)^{-1}$ is operator concave: operator convexity of the inverse follows by applying the inverse-quadratic variational identity to each vector. Thus (7.4), with $e = 1-\beta$, gives

$$Q_{\alpha,\mathrm{out}} \geq eQ_{\alpha,\mathrm{in}} + (1-e)Q_{\alpha,R}. \tag{8.4}$$

For positive a, b, weighted arithmetic–geometric mean gives $ea + (1-e)b \geq a^e b^{1-e}$. Taking minus logarithms in (8.4) therefore proves (8.3) for D_α. More explicitly, with $x = D_{\alpha,\mathrm{in}} - D_{\alpha,R} \geq 0$,

$$D_{\alpha,\mathrm{out}} - D_{\alpha,R} \leq -\frac{\log(1-e+ee^{-(1-\alpha)x})}{1-\alpha} \leq ex.$$

This is the logarithmic step needed for the conditional ratio.

Multiplying by $1-\alpha$ gives the same affine bound for $-\log Q_\alpha$. Taking the supremum over $0 < \alpha < 1$ yields

$$C_{Q,\mathrm{out}} \leq eC_{Q,\mathrm{in}} + (1-e)C_{Q,R},$$

since the supremum of the sum is at most the sum of the suprema. This proves the Chernoff upper bound. Subtraction and the product comparison in (7.2) give all the upper coefficient bounds.

8.2 Boundary passage without an endpoint-uniform assumption

For fixed interior α, matrix powers are continuous on PSD matrices, so (8.4) extends directly to arbitrary states. The same inequality also proves data processing when $e = 1$. In particular a finite input divergence has finite reference and output divergences.

For the Chernoff boundary, mix a pair with a common faithful state τ. Each Q_α is jointly concave. One way to see precisely the needed fact is to use (7.6): its variational inverse formula makes Q_s jointly convex in the state pair. Positive integration gives joint convexity of D_{f_α}, hence joint concavity of Q_α. Boundary continuity for fixed α extends that conclusion to PSD states. Therefore

$$Q_\alpha(\rho_\epsilon, \sigma_\epsilon) \geq (1-\epsilon)Q_\alpha(\rho,\sigma) + \epsilon.$$

Taking the infimum gives the corresponding lower bound on $\inf_\alpha Q_\alpha(\rho_\epsilon,\sigma_\epsilon)$. For every fixed interior order, continuity gives the reverse limsup bound; then take the infimum over that fixed order. Consequently

$$\inf_\alpha Q_\alpha(\rho_\epsilon,\sigma_\epsilon) \longrightarrow \inf_\alpha Q_\alpha(\rho,\sigma). \tag{8.5}$$

On the finite-Chernoff domain the limit is positive, so its negative logarithm converges. This argument does not assert uniform continuity of matrix powers in α at singular endpoints.

8.3 Local Rényi sharpness

At every faithful coincident pair, $-\log Q_\alpha = (1-Q_\alpha) + O((1-Q_\alpha)^2)$. Thus D_α and D_{f_α} have the same nonzero second-order coefficient under a traceless perturbation. Apply lemma 7.4 at fixed p, then send p to zero: the regular-generator factor cancels between input and output. This gives the lower bound in (8.1). For the uniform bit, Q_α is the average of its two block values. Its logarithm has the same second-order expansion, and the reference term is zero. This proves sharpness of the generalized information coefficient as well.

8.4 The optimized Chernoff converse

Lemma 8.2 (Uniform local order optimization)**.** *For* (5.1)*, fix* $0 < p < 1$*. Uniformly for* $0 \le \alpha \le 1$*,*

$$Q_\alpha(\omega_p + hH_p, \omega_p) = 1 - h^2 A_{\alpha,p} + O_p(|h|^3), \qquad A_{\alpha,p} = pq \sum_{i,j} |B_{ij}|^2 K_\alpha(qc_i, pv_j), \tag{8.6}$$

where

$$K_\alpha(a,b) = \frac{(a^\alpha - b^\alpha)(a^{1-\alpha} - b^{1-\alpha})}{(a-b)^2}, \qquad K_\alpha(a,a) = \frac{\alpha(1-\alpha)}{a}.$$

The maximum of $A_{\alpha,p}$ *is attained at* $\alpha = 1/2$*, and*

$$C_Q(\omega_p + hH_p, \omega_p) = h^2 A_{1/2,p} + O_p(|h|^3), \qquad \lim_{p \downarrow 0} \frac{A_{1/2,p}}{p} = q_C(B). \tag{8.7}$$

Proof. We first compute the coefficient. For a diagonal baseline with eigenvalues a, b and an off-diagonal entry hz, the second-order change of $\mathrm{Tr}(\omega + hH)^\alpha \omega^{1-\alpha}$ is $-|z|^2 K_\alpha(a,b)h^2$. This follows, for example, by the second divided difference of x^α: its contribution at the a entry is

$$a^{1-\alpha} \frac{b^\alpha - a^\alpha - \alpha a^{\alpha-1}(b-a)}{(b-a)^2} |z|^2,$$

and the analogous contribution at b has the opposite first-order linear term. Adding them gives $-K_\alpha(a,b)|z|^2$. The coincident-eigenvalue formula is its continuous limit. Summing the cross-sector entries $z = \sqrt{pq}\, B_{ij}$ proves (8.6).

The remainder is genuinely uniform at this fixed p. Choose $h_0 > 0$ so that $\omega_p + hH_p$ is bounded below by a positive multiple of the identity for $|h| \le h_0$. On this compact faithful set the map $(h, \alpha) \mapsto \exp(\alpha \log(\omega_p + hH_p))$ has continuous third derivatives in h for $\alpha \in [0,1]$. They are bounded on the compact parameter rectangle. Taylor's theorem therefore gives one $O_p(|h|^3)$ bound for all those orders.

For $a \neq b$, the numerator in K_α is

$$a + b - a^\alpha b^{1-\alpha} - a^{1-\alpha} b^\alpha \le a + b - 2\sqrt{ab}.$$

Equality holds at $\alpha = 1/2$ by arithmetic–geometric mean. The coincident case uses $\alpha(1-\alpha) \le 1/4$. Hence every nonnegative summand is maximized at one half, and $K_{1/2}(a,b) = (\sqrt{a} + \sqrt{b})^{-2}$. The uniform remainder now permits taking the infimum in (8.6) before the local limit in h. Taking a logarithm gives the first assertion in (8.7). Finally,

$$\frac{A_{1/2,p}}{p} = q \sum_{i,j} \frac{|B_{ij}|^2}{(\sqrt{qc_i} + \sqrt{pv_j})^2} \longrightarrow \sum_{i,j} \frac{|B_{ij}|^2}{c_i}.$$

This is a finite sum; its terms are bounded by $|B_{ij}|^2/c_i$. □

Completion of theorem 8.1. The same Chernoff lemma applies at the output, so its ordered local and rare-sector ratio is $q_{C'}(B')/q_C(B)$. For the binary auxiliary, the two compared direct-sum blocks have equal weights. Their Q_α is the average of the values at h and $-h$, uniformly in α; the coefficient in (8.6) is unchanged. The reference divergence vanishes exactly. Hence the same local Chernoff ratio proves the generalized information lower bound. The conditional ratio again contains the complete one. Use (3.14) and the already proved upper bounds. The construction has the same reference size as before. □

Corollary 8.3 (Faithful marginal constraints for the extensions)**.** *The complete coefficients in theorems 7.2 and 8.1 are unchanged when both input marginals are prescribed to a faithful state ω_A. The binary generalized-information witnesses may also have input marginal ω_A and be independent of each local subsystem. Reference dimension at most $m^2 + 2$ suffices.*

Proof. Use the padding in theorem 6.5, with one fixed $0 < a < \lambda_{\min}(\omega_A)$; since every input marginal is at most I_A, the residual density is positive. For modular D_f, both local divergences scale exactly by a. For D_α, the padded value is

$$-\frac{1}{1-\alpha}\log[1 - a + a\exp(-(1-\alpha)D_\alpha)] = aD_\alpha + O(D_\alpha^2)$$

at a coincident pair. For Chernoff it is $-\log(1 - a + ae^{-C_Q}) = aC_Q + O(C_Q^2)$, since an increasing affine map commutes with the infimum of Q_α. Consequently the local $h \to 0$ ratios in the sharpness construction are unchanged. The same block calculation applies to the uniform-bit comparison, whose reference term is exactly zero. Only one reference direction is added. □

This corollary concerns faithful prescribed marginals. For the nonfaithful, second-marginal-only restriction, the Umegaki support argument in theorem 6.5 must not be transferred to these divergences. For example, take the identity channel, $\sigma = |0\rangle\langle 0|$ and $\rho = \mathrm{diag}(1-t, t)$, $0 < t < 1$. The regular modular, interior-order Petz Rényi and Chernoff divergences of this pair are positive and finite, so the restricted ratio is one, whereas restricting the channel itself to $\operatorname{supp}\sigma$ gives a one-dimensional input and coefficient zero.

The uniformity used in lemma 8.2 is only at a fixed faithful ω_p. No Taylor remainder uniform in the rare-sector weight or in singular states has been used. In fact, an endpoint extension at Petz order zero is false.

Proposition 8.4 (A zero-order obstruction)**.** *For qubit depolarization with shrinkage $1/2$, all three coefficients defined from $D_0(\rho\|\sigma) = -\log \mathrm{Tr}(P_\rho\sigma)$ are zero. For every fixed $0 < \alpha < 1$, the three corresponding D_α coefficients equal $3/8$.*

Proof. For any reference, $\mathcal{N}_R(\rho) = \rho/2 + \rho_R \otimes I_2/4$ has support $\operatorname{supp}\rho_R \otimes \mathbb{C}^2$. Therefore $D_0(\mathcal{N}_R\rho\|\mathcal{N}_R\sigma) = D_0(\rho_R\|\sigma_R)$. Every conditional numerator is zero; equal-reference numerators also vanish. The same support identity with reference UR proves the generalized-information assertion. Positive finite input denominators exist, for example from a pure input versus $I_2/2$, or a uniform bit encoded in two orthogonal inputs. The interior value follows from theorems 8.1 and 9.2. □

Thus fixed-order exactness is not a uniform extremizer statement as the order approaches zero. Orders above one, sandwiched Rényi divergences, maximal divergences and hockey-stick divergences are not included. An explicit higher-order counterexample is recorded in appendix C.3.

9 Exact families and the mutual-information separation

We now evaluate the formula and compare the optimizations in table 2. Throughout this section, $\eta(\mathcal{N})$ denotes the common value in (2.7), or any of the complete, conditional-difference and generalized-information coefficients proved equal to it in theorems 7.2 and 8.1. It never denotes ordinary MI or ordinary unreferenced relative-entropy contraction.

9.1 Replacers and identity channels

For a replacer $\mathcal{R}_\tau(X) = \mathrm{Tr}(X)\tau$, $J = I_A \otimes \tau$. The feasible density $\sigma = \tau$ gives inverse objective one:

$$\mathrm{Tr}_B[(I \otimes \tau)J^+(I \otimes \tau)] = \mathrm{Tr}(\tau\tau^+\tau)\, I_A = I_A.$$

Since $M \geq 1$, $M = 1$, $\beta = 1$ and $\eta = 0$, including pure replacers. If $d > 1$, an identity or isometric channel has no complete output row in its rank-one Choi support, so $\beta = 0$ and $\eta = 1$. These examples also show why singularity alone does not determine whether contraction is strict.

9.2 Every finite classical channel

Let $W(y|x)$ be a stochastic matrix and regard it as the quantum channel that first measures x and then prepares the output basis state y. Define the common positive-output set $Y_0 = \{y : W(y|x) > 0 \text{ for all } x\}$.

Theorem 9.1 (The classical harmonic formula with quantum references)**.** *For every finite stochastic matrix,*

$$\boxed{\eta(W) = 1 - \min_{\theta \in \Delta_X} \sum_{y \in Y_0} \left(\sum_x \frac{\theta_x}{W(y|x)} \right)^{-1}.} \tag{9.1}$$

An empty sum is zero. The bound is sharp even when the channel input register is required to be classical; the untouched reference can be quantum.

Proof. The Choi operator is diagonal with entries $W(y|x)$. Its feasible centers are supported on the span of Y_0. For a possibly nondiagonal density σ on that span, the xth diagonal entry of its inverse form is

$$\sum_{y \in Y_0} \frac{(\sigma^2)_{yy}}{W(y|x)} \geq \sum_{y \in Y_0} \frac{\sigma_{yy}^2}{W(y|x)}.$$

The remaining input matrix is diagonal. Removing off-diagonal center entries cannot increase its norm, so

$$M(W) = \min_{\substack{s_y \geq 0 \\ \sum_{y \in Y_0} s_y = 1}} \max_x \sum_{y \in Y_0} \frac{s_y^2}{W(y|x)}. \tag{9.2}$$

If Y_0 is empty the program is infeasible. Otherwise, convex minimax and scalar weighted Cauchy–Schwarz give

$$M = \max_{\theta \in \Delta_X} \left[\sum_{y \in Y_0} \left(\sum_x \frac{\theta_x}{W(y|x)} \right)^{-1} \right]^{-1}.$$

Taking the reciprocal proves (9.1).

To justify sharpness with a classical input, reconsider the minimax proof of lemma 3.1. Both $\mathbf{A}$ and $\mathbf{G}$ are block diagonal in x. Dephasing C in that input basis preserves every minimax objective, and it preserves faithfulness. For the weighted projection in (3.3), the Riesz representative has the form $C(\Phi(F) - aI)$ at its minimizing center; it is also block diagonal in x and its blockwise trace is zero by the center's first-order optimality condition. Thus the witnessing B can be chosen block diagonal in x. Choose τ diagonal in that basis in (5.1). The coherent states are then classical on the channel input, while their off-diagonal entries occur in the reference. The converse arguments for all the stated divergences retain this property. □

For the k-symbol symmetric channel, with diagonal $a > 0$, off-diagonal $b > 0$ and $a+(k-1)b = 1$, permutation symmetry and convexity in (9.2) make $s_y = 1/k$ optimal. Hence

$$\eta(W) = 1 - \frac{k^2}{1/a + (k-1)/b}. \tag{9.3}$$

This differs from its ordinary classical KL coefficient

$$\eta_{\mathrm{KL,class}}(W) = \frac{(a-b)^2}{a+b}, \tag{9.4}$$

whose full elementary proof is in appendix A. The general binary-input reduction for ordinary classical SDPI is established in [13, Theorem 1]; the classical formula is not claimed as a new method here.

Channel	Parameters	Ordinary classical KL	Quantum-reference η
Symmetric ternary	$a = 3/5,\ b = 1/5$	$1/5$	$8/35$
Symmetric ternary	$a = 9/10,\ b = 1/20$	$289/380$	$289/370$
k-symbol exclusion	$a = 0,\ b = 1/(k-1)$	$1/(k-1)$	1

Table 3: A classical channel can retain substantially more distinguishability in correlations with a quantum reference. The ordinary column restricts the input pair itself to probability distributions.

The ternary triple $(.9, .05, .05)$ and its permutations is discussed in [2, Section 9.4] for multihypothesis Chernoff quantities. Those testing exponents are different from both columns in table 3. Neither the symmetric channel nor established classical testing formulas are being relabelled as a new construction. The reference-assisted value follows from (9.1).

For the exclusion channel on $k \geq 3$ symbols, every output has a zero transition, so Y_0 is empty and $\eta = 1$; its ordinary KL coefficient is $1/(k-1)$. The separation also persists without zero entries. Choose $a = k^{-3}$ and $b = (1-a)/(k-1)$. Then

$$\eta(W) > 1 - \frac{1}{k}, \qquad \eta_{\mathrm{KL,class}}(W) < b < \frac{1}{k-1}.$$

The first inequality follows from $1/a = k^3$ in (9.3); the second follows from $0 < a < b$ in (9.4). The ratio of the two coefficients is therefore unbounded as the alphabet grows.

9.3 Depolarizing channels in all dimensions

For $d \geq 2$ let

$$\mathcal{D}_\lambda(X) = \lambda X + (1-\lambda)\,\mathrm{Tr}(X)\frac{I_d}{d}, \qquad -\frac{1}{d^2-1} \leq \lambda \leq 1.$$

This is its full CP interval.

Theorem 9.2 (Exact complete depolarizing coefficient)**.** *On the full CP interval,*

$$\boxed{\eta(\mathcal{D}_\lambda) = \frac{(d^2-1)\lambda^2}{1+(d^2-2)\lambda}.} \tag{9.5}$$

In dimension one the channel is a replacer and its coefficient is zero.

Proof. In the interior of the interval, $J = \lambda|\Omega\rangle\langle\Omega| + (1-\lambda)I_{d^2}/d$ is positive definite, where $|\Omega\rangle = \sum_i |ii\rangle$. Its eigenvalues are

$$a = \frac{1+(d^2-1)\lambda}{d} \quad \text{once}, \qquad b = \frac{1-\lambda}{d} \quad d^2-1 \text{ times}.$$

It is invariant under $\overline{U}\otimes U$. Conjugating a feasible center by U leaves the inverse objective unchanged. That objective is convex: the difference between the weighted average of the two quadratic matrices and the matrix of the averaged center is a positive partial trace of a square. Averaging over the d^2 Weyl conjugations therefore makes I_d/d optimal. The inverse Choi marginal is scalar, giving

$$M = \frac{\operatorname{Tr} J^{-1}}{d^3} = \frac{1}{d^2}\left(\frac{1}{1+(d^2-1)\lambda} + \frac{d^2-1}{1-\lambda}\right) = \frac{1+(d^2-2)\lambda}{(1-\lambda)(1+(d^2-1)\lambda)}.$$

Substitution into $1-1/M$ proves (9.5).

At the negative CP endpoint the kernel of J contains the maximally entangled vector $|\Omega\rangle$. A feasible density would have $(I\otimes\sigma)|\Omega\rangle = 0$, forcing $\sigma = 0$, a contradiction. At $\lambda = 1$, the kernel contains a maximally entangled vector orthogonal to $|\Omega\rangle$, giving the same contradiction. Thus the coefficient is one at both singular endpoints, exactly the value of (9.5). At $\lambda = 0$ the channel is a replacer. □

9.4 Pinching from a classical reference-assisted channel

The next lemma supplies the coherence estimate needed for ordinary MI. It transfers the already proved classical coefficient through a flagged unitary orbit; it requires no separate conditional-metric reconstruction.

Lemma 9.3 (Finite-twirl coherence bound)**.** *Let $\mathcal{P} = N^{-1}\sum_{g\in G}\operatorname{Ad}U_g$ be the uniform twirl of a group of $N \geq 2$ conjugation actions. Put $\Gamma_\theta = \theta\,\mathrm{id} + (1-\theta)\mathcal{P}$. For $-1/(N-1) \leq \theta \leq 1$ and any state ρ, including with an arbitrary untouched reference,*

$$D(\Gamma_\theta\rho\|\mathcal{P}\rho) \leq c_N(\theta)D(\rho\|\mathcal{P}\rho), \qquad c_N(\theta) = \frac{(N-1)\theta^2}{1+(N-2)\theta}. \tag{9.6}$$

Proof. The N-symbol uniform-reset channel $K_{\theta,N}(y|x) = \theta\delta_{xy} + (1-\theta)/N$ is stochastic on exactly the stated interval. By (9.3), including its zero-entry endpoints, its complete coefficient is $c_N(\theta)$. Form the classical-quantum states

$$\Omega = \frac{1}{N}\sum_g |g\rangle\langle g| \otimes U_g\rho U_g^*, \qquad \Xi = \frac{I_G}{N}\otimes\mathcal{P}\rho.$$

Their untouched-system marginals both equal $\mathcal{P}\rho$. Group averaging implies $U_g(\mathcal{P}\rho)U_g^* = \mathcal{P}\rho$. After $K_{\theta,N}$ acts on the classical register, the gth normalized block of Ω becomes $U_g\Gamma_\theta(\rho)U_g^*$, and Ξ is unchanged. Orthogonal direct sums and unitary invariance therefore give

$$D(\Omega\|\Xi) = D(\rho\|\mathcal{P}\rho), \quad D((K_{\theta,N}\otimes\mathrm{id})\Omega\|\Xi) = D(\Gamma_\theta\rho\|\mathcal{P}\rho).$$

The complete classical coefficient proves the inequality. These entropies are finite since $\rho \preceq N\mathcal{P}\rho$; the identity action appears in the twirl. Tensoring the whole construction with a reference changes neither the argument nor the constant. □

9.5 Ordinary MI: exact for qubits, bounds in higher dimension

Theorem 9.4 (Ordinary quantum MI is a distinct coefficient)**.** *For qubit depolarization on* $-1/3 \le \lambda \le 1$*,*

$$\eta_D^{\mathrm{ord}}(\mathcal{D}_\lambda) = \lambda^2, \qquad \boxed{\eta_{\mathrm{MI}}(\mathcal{D}_\lambda) = \frac{2\lambda^2}{1+\lambda}.} \tag{9.7}$$

A reference qubit suffices to approach the MI value. For every allowed $\lambda \notin \{0, 1\}$*, it is strictly smaller than* $\eta_D^{\mathrm{c}}(\mathcal{D}_\lambda) = 3\lambda^2/(1+2\lambda)$*.*

More generally, for $d \ge 2$ *on the full CP interval,*

$$\frac{d\lambda^2}{1+(d-1)\lambda} \le \eta_{\mathrm{MI}}(\mathcal{D}_\lambda) \le U_d(\lambda) := \frac{d(d-1)\lambda^2}{1+(d^2-d-1)\lambda}. \tag{9.8}$$

For $\lambda \notin \{0, 1\}$ *the upper bound is strictly below* (9.5)*. No exact higher-dimensional MI formula is asserted.*

Proof. Fix ρ_{AR} and diagonalize ρ_A. Let Δ dephase A in that basis. Dephasing preserves both marginals, so the pinching entropy identity gives

$$I(A;R)_\rho = I(A;R)_{\Delta\rho} + D(\rho\|\Delta\rho). \tag{9.9}$$

The identity follows by subtracting the two entropy expressions and using $\operatorname{Tr}\rho\log\Delta\rho = \operatorname{Tr}\Delta\rho\log\Delta\rho$. Depolarization commutes with Δ, and the output marginal remains diagonal in the same basis, even for negative λ. Thus (9.9) also holds at the output.

On states diagonal in A, depolarization is the d-symbol uniform-reset channel. Apply theorem 9.1 to the pair $\Delta\rho$ and $\rho_A \otimes \rho_R$, whose reference marginals agree. The diagonal information contracts by at most $c_d(\lambda)$.

For the coherence term let S cyclically shift the chosen basis and put

$$k = \frac{1+(d-1)\lambda}{d}, \qquad \theta = \lambda/k, \qquad \Gamma_\theta = \theta\,\mathrm{id} + (1-\theta)\Delta.$$

Writing physical matrix blocks verifies the two positive mixtures

$$\mathcal{D}_\lambda\rho = k\Gamma_\theta\rho + \frac{1-\lambda}{d}\sum_{j=1}^{d-1} S^j\Delta\rho S^{-j},$$

$$\Delta\mathcal{D}_\lambda\rho = k\Delta\rho + \frac{1-\lambda}{d}\sum_{j=1}^{d-1} S^j\Delta\rho S^{-j}.$$

The weights add to one. Across the full CP interval, $k \ge 1/(d+1) > 0$ and $-1/(d-1) \le \theta \le 1$, so Γ_θ is CPTP by its d-element phase-twirl representation. Joint convexity of relative entropy, followed by lemma 9.3 for that phase twirl, gives

$$D(\mathcal{D}_\lambda\rho\|\Delta\mathcal{D}_\lambda\rho) \le kD(\Gamma_\theta\rho\|\Delta\rho) \le kc_d(\theta)D(\rho\|\Delta\rho) = U_d(\lambda)D(\rho\|\Delta\rho).$$

The joint-convexity and boundary facts used here are proved in appendix B.2. Since $U_d(\lambda) \ge c_d(\lambda)$, summing the two nonnegative terms in (9.9) proves the upper bound. For $d = 2$ the diagonal bound is λ^2, while $\Delta = (\mathrm{id} + \mathrm{Ad}\,Z)/2$ and $kc_2(\theta) = k\theta^2 = 2\lambda^2/(1+\lambda)$. This explicitly gives the binary-cq and binary-pinching foundations.

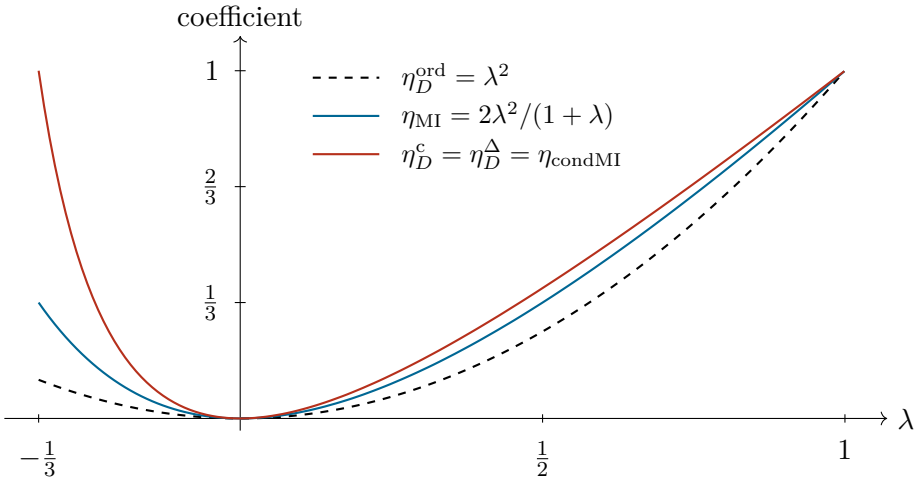


Figure 2: Three distinct qubit depolarizing contraction coefficients over the full CP interval. At $\lambda = 1/2$ their values are $1/4, 1/3, 3/8$. At $\lambda = -1/3$ they are $1/9, 1/3, 1$.

For the matching qubit lower bound, use $0 < p, s < 1$ and

$$\tau = \mathrm{diag}(p, 1-p)_A \otimes \mathrm{diag}(1-s, s)_R, \qquad \rho_h = \tau + h(|00\rangle\langle 11| + |11\rangle\langle 00|).$$

It is faithful for $h^2 < p(1-p)s(1-s)$. Both marginals equal those of τ, so its MI is exactly $D(\rho_h\|\tau)$. The same is true at the output. At fixed p, s, the local ratio as $h \to 0$ is

$$\lambda^2 \frac{L(p'(1-s), (1-p')s)}{L(p(1-s), (1-p)s)}, \quad p' = \lambda p + (1-\lambda)/2, \quad L(a,b) = \frac{\log a - \log b}{a-b}.$$

Sending $s \downarrow 0$ gives $\lambda^2 p/p'$, and then $p \uparrow 1$ gives $2\lambda^2/(1+\lambda)$. For any prescribed tolerance choose p, then s, and finally a sufficiently small nonzero h. Each witness is a finite faithful state, and no remainder uniform in p, s is used.

The same family in dimension d, supported in the tangent on two physical levels, gives the lower bound in (9.8). Choose a faithful diagonal physical marginal with first population p and distribute $1-p$ positively over the other levels. Use the same off-diagonal perturbation between the first physical level with the common reference state and a second physical level with the rare reference state. The ordered local and rare-reference limits give $\lambda^2 p/[\lambda p + (1-\lambda)/d]$. Let $p \uparrow 1$. Positivity is ensured first by fixing the populations and then choosing h sufficiently small.

For $d = 2$ the two bounds in (9.8) coincide. The ordinary qubit relative-entropy coefficient is proved independently by the Bloch-metric calculation in appendix B.3. Finally, direct subtraction gives

$$\eta_D^{\mathrm{c}}(\mathcal{D}_\lambda) - U_d(\lambda) = \frac{(d-1)\lambda^2(1-\lambda)}{[1+(d^2-2)\lambda][1+(d^2-d-1)\lambda]}.$$

Both denominators are positive throughout the CP interval. This proves the strict separation. The constant and identity endpoints are immediate. □

10 Orthogonal-unitary noise and a scoped mixing application

For a specified error basis, the finite Choi optimization becomes a scalar inverse-probability formula. Its dependence on rare errors explains why short noise words and spectral gaps answer different

quantitative questions.

Theorem 10.1 (Orthogonal-unitary inverse probabilities)**.** *Let $\{U_g : g = 1, \dots, d^2\}$ be an orthogonal unitary basis, $\operatorname{Tr} U_g^* U_h = d\delta_{gh}$, and let $\mathcal{N}_p(X) = \sum_g p_g U_g X U_g^*$. If all probabilities are positive,*

$$M(\mathcal{N}_p) = \frac{1}{d^4} \sum_g \frac{1}{p_g}, \qquad \beta(\mathcal{N}_p) = \frac{d^4}{\sum_g p_g^{-1}}. \tag{10.1}$$

If a probability is zero, $M = \infty$ and $\beta = 0$.

Proof. The normalized vectors $|u_g\rangle = |U_g\rangle\!\rangle/\sqrt{d}$ form an orthonormal maximally entangled basis, and $J = d\sum_g p_g |u_g\rangle\langle u_g|$. When all $p_g > 0$,

$$\operatorname{Tr}_A J^{-1} = \operatorname{Tr}_B J^{-1} = cI_d, \qquad c = d^{-2} \sum_g p_g^{-1}.$$

For every density σ, its positive inverse form T_σ satisfies

$$\|T_\sigma\|_\infty \geq \frac{\operatorname{Tr} T_\sigma}{d} = \frac{c}{d} \operatorname{Tr} \sigma^2 \geq \frac{c}{d^2}.$$

The center I_d/d attains both inequalities, proving (10.1). If $p_g = 0$, feasibility would force $(I \otimes \sigma)|u_g\rangle = 0$. Full Schmidt rank forces $\sigma = 0$, which is not a density. □

Theorem 10.2 (Shortest-word onset for a fixed Weyl noise family)**.** *Let the orthogonal basis be a projective Weyl representation of a finite label group G, $|G| = d^2$, and consider*

$$\mathcal{L} = \sum_{s \in S} \gamma_s (\operatorname{Ad} U_s - \mathrm{id}), \qquad T_t = e^{t\mathcal{L}}, \qquad \gamma_s > 0.$$

Suppose S generates G. Let $\ell(g)$ be the shortest directed word length reaching g, put $r = \max_g \ell(g)$, and set

$$c_e = 1, \qquad c_g = \frac{1}{\ell(g)!} \sum_{s_1 \cdots s_{\ell(g)} = g} \prod_{j=1}^{\ell(g)} \gamma_{s_j} \quad (g \neq e).$$

At fixed group and fixed rates, as $t \downarrow 0$,

$$\beta(T_t) = \frac{d^4}{\sum_{\ell(g)=r} c_g^{-1}} t^r + O(t^{r+1}). \tag{10.2}$$

If the jumps generate only a proper subgroup, $\beta(T_t) = 0$ for every finite $t > 0$. Given unit-cost group operations, all shortest lengths and their leading coefficients require $O(|G||S|)$ arithmetic operations.

Proof. Put $\Gamma = \sum_s \gamma_s$. The compound-Poisson expansion is

$$p_t(g) = e^{-\Gamma t} \sum_{k \geq 0} \frac{t^k}{k!} \sum_{s_1 \cdots s_k = g} \prod_{j=1}^{k} \gamma_{s_j}.$$

Projective phases disappear under conjugation. For each fixed g, $p_t(g) = c_g t^{\ell(g)} + O(t^{\ell(g)+1})$. Substitute into (10.1); the reciprocal sum is dominated by all labels at maximal distance r. Inverting that finite positive leading sum proves (10.2). If a label is unreachable, its probability is zero at all finite times and theorem 10.1 applies.

Breadth-first search gives the directed lengths. On the resulting acyclic shortest-path levels, sum products of rates over the predecessors to obtain the shortest-word weights, then divide at level k by $k!$. Each edge is used a bounded number of times. This proves the stated arithmetic count; it is not a bound uniform in dimension or a claim about acquisition of the rates. □

For n qubits the full error basis has 4^n labels. The finite-group algorithm is therefore not polynomial in the qubit count. The asymptotic remainder in (10.2) also concerns a fixed family; it cannot be used without control when G changes with n.

10.1 Two independent Pauli errors

For one qubit with X and Z jumps at rates $a, b > 0$, the two parity bits are independent, with odd probabilities $(1 - e^{-2at})/2$ and $(1 - e^{-2bt})/2$. Factoring their inverse-probability sum gives the exact identities

$$\beta(T_t) = (1 - e^{-4at})(1 - e^{-4bt}), \qquad \alpha_+(T_t) = (1 - e^{-2at})(1 - e^{-2bt}). \tag{10.3}$$

The second follows from $\alpha_+ = 4 \min_g p_t(g)$, as proved in theorem 6.4. In particular $\beta(T_t) = 16ab\,t^2 + O(t^3)$. The nonidentity Pauli eigenoperators have generator eigenvalues $-2a, -2b, -2(a+b)$, so the spectral gap is $2\min(a, b) > 0$. Nevertheless

$$\lim_{t \downarrow 0} \frac{-\log \eta(T_t)}{t} = 0.$$

There is no $\kappa > 0$ with $\eta(T_t) \le e^{-\kappa t}$ uniformly from time zero. This is an all-pairs reference-assisted statement and is compatible with positive fixed-equilibrium entropy rates in other inequalities [5].

10.2 An exact tensor-product transition

For n independent qubits with equal X, Z rates $\gamma > 0$, theorem 6.3 and (10.3) give, at every $t \ge 0$,

$$\eta_n(t) = 1 - (1 - e^{-4\gamma t})^{2n}, \tag{10.4}$$

$$T_n(\epsilon) = -\frac{1}{4\gamma} \log\left[1 - (1 - \epsilon)^{1/(2n)}\right], \qquad 0 < \epsilon < 1. \tag{10.5}$$

Here $T_n(\epsilon)$ is the first time $\eta_n(t) \le \epsilon$. At $t = (\log(2n) + c)/(4\gamma)$,

$$\eta_n(t) \longrightarrow 1 - \exp(-e^{-c}).$$

These follow directly by solving the scalar equation and taking $(1 - e^{-c}/(2n))^{2n} \to e^{-e^{-c}}$.

The positive CP-replacer bound $\eta_n(t) \le 1 - \alpha_{+,n}(t)$ uses $\alpha_{+,n}(t) = (1 - e^{-2\gamma t})^{2n}$. For a fixed threshold it reaches its transition near $\log(2n)/(2\gamma)$, whereas (10.5) has center $\log(2n)/(4\gamma)$. This factor-two comparison is to that specific Doeblin bound. Tensor-product quantum mixing cutoffs and the limitations of spectral-gap estimates are established in [11]; we claim neither a new general cutoff theorem nor superiority over every available mixing method.

11 Discontinuity and the obstruction to uniform acquisition

An exact finite formula does not imply that its value is uniformly learnable from experiments on an unknown channel. The support condition makes this distinction sharp, even for one qubit.

Theorem 11.1 (Jump at pure reset)**.** *Let $\mathcal{A}_\gamma$ be qubit amplitude damping with Kraus operators*

$$K_0 = \mathrm{diag}(1, \sqrt{1 - \gamma}), \qquad K_1 = \sqrt{\gamma}\,|0\rangle\langle 1|, \qquad 0 \le \gamma \le 1.$$

For every $0 \le \gamma < 1$, its complete and conditional Umegaki coefficients and its conditional-MI coefficient are one. The analogous three coefficients for every divergence in theorems 7.2 and 8.1 are also one. At $\gamma = 1$ all these coefficients are zero. Nevertheless,

$$\|\mathcal{A}_\gamma - \mathcal{A}_1\|_\diamond \le 2\sqrt{2(1 - \sqrt{\gamma})} \longrightarrow 0. \tag{11.1}$$

Proof. There is a direct coherent witness, independent of solving the Choi optimization. In (5.1) take $R_0 = \mathbb{C}$, $C = \operatorname{diag}(1-r, r)$, $0 < r < 1$, $B = |1\rangle\langle 0|$ and $\tau = I_2/2$. Then $\operatorname{Tr} B = 0$ and $q_C(B) = 1/r$. Writing $s = 1 - \gamma > 0$ gives

$$C' = \operatorname{diag}(1 - sr, sr), \qquad B' = \sqrt{s}\,|1\rangle\langle 0|, \qquad q_{C'}(B') = 1/r.$$

The coherent states are positive when $h^2 < r/2$, independently of their rare-sector weight, and both marginals are exactly fixed. For Umegaki entropy their input and output Hessians are

$$2pq\,L(qr, p/2), \qquad 2pqs\,L(qsr, p(1+\gamma)/2), \quad L(a,b) = \frac{\log a - \log b}{a - b}.$$

Both, divided by $2p\log(1/p)$, tend to $1/r$. The ordered perturbation and rare-sector limits thus give ratio one. The uniform bit of section 5 gives the same CMI ratio. For regular Petz and logarithmic divergences, lemmas 7.4 and 8.2 give the same ratio from the two equal rectangular forms. Data processing bounds each coefficient above by one; the conditional numerator is at most its input counterpart by subtraction. Thus all these values are exactly one. At $\gamma = 1$ the channel is the pure replacer, so every output divergence equals the reference divergence and each coefficient is zero.

For the norm bound use a common environment and isometries

$$V_\gamma|0\rangle = |00\rangle, \qquad V_\gamma|1\rangle = \sqrt{1-\gamma}|10\rangle + \sqrt{\gamma}|01\rangle.$$

Their difference vanishes on $|0\rangle$ and has squared norm $2(1 - \sqrt{\gamma})$ on $|1\rangle$. For pure inputs with any reference, trace distance of the two isometric outputs is at most twice their vector distance. Convexity and partial-trace contractivity give $\|\mathcal{A}_\gamma - \mathcal{A}_1\|_\diamond \le 2\|V_\gamma - V_1\|$, proving (11.1). □

The lower witness uses a shrinking perturbation and a rare reference sector. Its ratio remains close to one while its absolute information becomes arbitrarily small. The theorem does not supply a finite communication rate near complete damping.

Theorem 11.2 (No uniformly bounded-use estimator)**.** *Fix $0 < \epsilon < 1/2$. There is no estimator that, for every unknown qubit channel, estimates its complete Umegaki coefficient to additive error ϵ with probability at least $2/3$, while using at most a uniformly bounded number of channel calls.*

The access model permits arbitrary finite ancillas, entangled inputs, adaptive CPTP processing, intermediate measurements and classical feedforward, and a final measurement. Calls are ordinary uses of the unknown CPTP map. Access to a chosen dilation, its inverse, or a controlled oracle is not provided. The same impossibility holds for each of the other fixed coefficients in theorem 11.1.

Proof. Suppose the worst-case call bound is N. Pad protocols with fewer calls by ignored calls if needed. For two channels, replace one call at a time in the N-round adaptive protocol. At each replacement the trace norm changes by at most their diamond-norm distance; subsequent CPTP processing cannot increase it. Thus the final states have trace distance, with the factor $1/2$ convention, at most

$$\frac{N}{2}\|\mathcal{A}_\gamma - \mathcal{A}_1\|_\diamond.$$

Intermediate instruments and classical records are part of the retained system, so this argument includes adaptive strategies. Any final measurement has total-variation distance at most this value.

The exact coefficients are one for $\gamma < 1$ and zero for $\gamma = 1$. Because $\epsilon < 1/2$, thresholding a successful estimate at $1/2$ distinguishes these channels with success at least $2/3$ under either hypothesis. The event that the estimate exceeds $1/2$ must therefore have probabilities differing by at least $1/3$. Choose $\gamma < 1$ sufficiently close to one that the displayed distance is smaller than $1/3$, using (11.1). This contradiction proves the theorem, including $N = 0$. □

This is an information-access obstruction, independent of computational power. It does not preclude evaluation from exact supplied Choi entries, or an acquisition theorem under explicit support and spectral-margin promises. It also does not preclude a pointwise procedure whose required number of calls diverges near the discontinuity.

12 Scope and further questions

The main operation established here is exact evaluation of a specified reference-assisted contraction problem from a finite channel description. The complete and reference-subtracted Umegaki coefficients are equal for every finite channel, and their value is the finite Choi program. The explicit conditional-MI proof uses a separate conditioning memory. For depolarizing channels the ordinary-MI coefficient is strictly smaller at every nonconstant, nonidentity parameter, with an exact value for qubits and proved bounds in higher dimension. These conclusions answer the two comparisons posed for depolarization in [9, Section 3.3, equation (3.66)]: the first is equality, and the second is strict away from the two trivial parameters. They do not compute ordinary MI for an arbitrary channel.

The program has a block constraint of order $2dm$, an $m \times m$ density variable, and a Hermitian $dm \times dm$ auxiliary variable. It eliminates unbounded reference dimension from evaluation. This is a dense matrix input statement. It does not bypass the cost of obtaining Choi entries from a circuit, a many-body description or experiments. Nor is a finite SDP an assertion of uniformly conditioned or strongly polynomial numerical evaluation. Exact support decisions and the difference between strict feasibility and a nearby singular channel remain consequential.

The regular modular Petz extension has finite endpoint value and recession slope as genuine hypotheses. Fixed interior Petz Rényi orders and Chernoff divergence require their own logarithmic and optimization arguments. No assertion has been made for sandwiched, maximal or hockey-stick divergences, or for general orders above one. The reference dimensions bound the witnesses needed for a supremum; they give neither a lower bound on the information in those witnesses nor a uniform precision requirement for constructing them.

The signed lift reconstructs observable means with weighted-square control. It is not required to preserve positive observables, and it has not been converted into a physical postprocessing channel. In particular the scalar self-coefficient alone supplies no comparison between a receiver and its complement. Capacity and recovery conclusions would require additional channel-order hypotheses, of the kind studied in [17, 8].

Several precise questions remain. What support and spectral promises make acquisition from noisy channel data stable with useful costs? What remains of exact sharpness when absolute input information and preparation precision are constrained? Can the exact ordinary-MI coefficient be found in dimension greater than two? For which additional divergences does the positive-center erasure comparison give the sharp linear coefficient? General noncommuting noise-onset prefactors and succinct-generator complexity involve further structures beyond the fixed-Weyl application proved here.

The distinction between exact evaluation, robust acquisition and a useful physical protocol is therefore retained throughout. Each is a separate mathematical operation.

Acknowledgements and AI use

The mathematical ideas, constructions, proofs, and manuscript were developed by GPT-6 Astra, with additional AI assistance for review. Human input was minimal and consisted primarily of setting the research objectives and providing occasional feedback.

A The ordinary symmetric classical coefficient

This appendix proves (9.4) independently of the reference-assisted theorem. Let $k \geq 2$, $a \geq 0$, $b > 0$ and $a + (k-1)b = 1$, and put $c = a - b$. For an interior probability vector p and a real tangent v with $\sum_i v_i = 0$, the input and output KL Hessians are

$$\sum_i \frac{v_i^2}{p_i}, \qquad c^2 \sum_i \frac{v_i^2}{b + cp_i}.$$

If $c = 0$ the channel is constant. Otherwise it suffices to prove

$$\sum_i d_i v_i^2 \geq 0, \qquad d_i = \frac{1}{p_i} - \frac{a+b}{b+cp_i} = \frac{b(1-2p_i)}{p_i(b+cp_i)}. \tag{A.1}$$

All denominators are positive at an interior prior, including when $a = 0$. If every $p_i \leq 1/2$, each summand is nonnegative. Otherwise exactly one coordinate, call it p_0, exceeds $1/2$. Put $q = 1 - p_0 < 1/2$. For the other coordinates $d_i > 0$, and weighted Cauchy–Schwarz gives

$$\sum_{i \neq 0} d_i v_i^2 \geq \frac{v_0^2}{\sum_{i \neq 0} d_i^{-1}}.$$

For $0 < t < 1/2$, set $h(t) = t(b + ct)/[b(1-2t)]$. The function $h(t)/t$ is increasing, because its derivative is $(a+b)/[b(1-2t)^2] > 0$. Consequently

$$\sum_{i \neq 0} d_i^{-1} = \sum_{i \neq 0} h(p_i) \leq h(q).$$

Moreover $h(q) \leq -1/d_0$. Indeed, after multiplying by the common positive denominator $b(2p_0 - 1)$, this is $q(b + cq) \leq p_0(b + cp_0)$; the difference is $(2p_0 - 1)(b + c) = (2p_0 - 1)a \geq 0$. Thus the last weighted sum is at least $-d_0 v_0^2$, proving (A.1).

Integrate the Hessian comparison along the segment between two positive distributions to obtain the KL upper bound $c^2/(a+b)$. Common faithful regularization and lemma 4.2 extend it to all finite-KL pairs. For sharpness, use the prior placing probability $1/2$ on each of two symbols and perturb those two probabilities oppositely. The two relevant output probabilities are both $(a+b)/2$. Its Hessian ratio is exactly $c^2/(a+b)$, and the finite-pair ratios approach it. This proves (9.4), including the exclusion case $a = 0$.

For the associated measure-and-prepare quantum channel, its ordinary unreferenced quantum coefficient has the same value. Dephasing the input pair cannot increase relative entropy; applying the stochastic channel then gives the classical upper bound. Diagonal input pairs give the reverse bound.

B Information identities and metric facts

B.1 Product-reference contraction equals ordinary MI contraction

We prove the identity used in proposition 2.4, with credit to the corresponding result in [9, Corollary 3]. Put $\kappa = \eta_{\mathrm{MI}}(\mathcal{N})$. First $\eta_D^{\mathrm{ord}} \le \kappa$. Given a finite-relative-entropy pair ρ, σ, a classical bit selects ρ with probability a and σ with probability $1-a$. Its average is $\mu_a = a\rho + (1-a)\sigma$, and

$$I(U;A) = aD(\rho\|\mu_a) + (1-a)D(\sigma\|\mu_a).$$

On $\operatorname{supp}\sigma$, the state σ is faithful and ρ is supported there. As $a \downarrow 0$, $D(\rho\|\mu_a) \to D(\rho\|\sigma)$ and $D(\sigma\|\mu_a) = O(a^2)$. The first assertion uses continuity of the logarithm on that support; the second is its Taylor expansion with zero first derivative. The identical assertions hold after the channel on $\operatorname{supp}\mathcal{N}(\sigma)$. Therefore the MI ratio tends to the relative-entropy ratio, proving $\eta_D^{\mathrm{ord}} \le \kappa$.

For any product-reference comparison with finite input entropy,

$$D(\rho_{RA}\|\rho_R \otimes \sigma_A) = I(R;A)_\rho + D(\rho_A\|\sigma_A).$$

At the output the two terms contract by at most κ and η_D^{ord}, respectively. Thus their sum contracts by at most κ, proving $\eta_D^{\mathrm{prod}} \le \kappa$. If the input MI is zero its output MI is also zero, since the state is a product, so this argument includes zero-information summands. Choosing $\sigma_A = \rho_A$ gives the reverse inequality $\eta_D^{\mathrm{prod}} \ge \kappa$. Finally these product-reference pairs have the same reference marginal, so $\eta_D^{\mathrm{prod}} \le \eta_D^{\mathrm{c}}$. Together with theorem 2.3 this proves (2.10).

B.2 Joint convexity from the metric representation

For faithful ω and Hermitian H, each integrand in (4.6) is jointly convex in (ω, H). Indeed, by inverse quadratic duality it is the supremum over Y of

$$2\operatorname{Re}\operatorname{Tr} Y^*H - \operatorname{Tr} Y^*[(1-u)L_\omega + uR_\omega]Y,$$

which is affine in that pair. Integration proves joint convexity of $g_\omega(H)$. The affine-path formula (4.9) then proves joint convexity of $D(\rho\|\sigma)$ for faithful pairs. For pairs with finite relative entropy, mix every state with the same faithful state and apply lemma 4.2. If a positive-weight term on the right is infinite, the inequality is automatic. This proves precisely the convexity used in theorem 9.4, including the required boundary cases.

For binary pinching there is also a direct coherence proof. Write $\Delta = (\mathrm{id} + \operatorname{Ad} Z)/2$ with $Z^2 = I$, and set $\sigma = \Delta\rho$, $H = \rho - \sigma$. For faithful ρ, the function $t \mapsto g_{\sigma+tH}(H)$ is convex by the preceding argument and even because $Z\sigma Z = \sigma$, $ZHZ = -H$. It is therefore nondecreasing for $0 \le t \le 1$. For $0 \le \theta \le 1$, the affine-path formula gives

$$D(\sigma + \theta H\|\sigma) = \theta^2 \int_0^1 (1-s) g_{\sigma+\theta sH}(H)\,\mathrm{d}s \le \theta^2 D(\rho\|\Delta\rho).$$

For negative θ, conjugation by Z reduces the assertion to $|\theta|$. Common faithful regularization and the entropy identity $D(\rho\|\Delta\rho) = S(\Delta\rho) - S(\rho)$ handle arbitrary states. This is an independent check of the two-sector case of lemma 9.3.

B.3 Ordinary qubit depolarizing contraction

Represent a qubit state by $\rho = (I + \mathbf{r}\cdot\boldsymbol{\sigma})/2$, $r = |\mathbf{r}| < 1$. On the trace-one affine space the entropy potential is

$$F(r) = \frac{1+r}{2}\log\frac{1+r}{2} + \frac{1-r}{2}\log\frac{1-r}{2}.$$

For a tangent $H = (\mathbf{v}\cdot\boldsymbol{\sigma})/2$, decompose $\mathbf{v}$ into components parallel and perpendicular to $\mathbf{r}$. Differentiating this radial function gives

$$g_\rho(H) = \frac{|\mathbf{v}_\parallel|^2}{1-r^2} + \frac{\operatorname{arctanh} r}{r}|\mathbf{v}_\perp|^2.$$

At $r = 0$ both coefficients have limit one. Each coefficient is nondecreasing in r: for the second use $\operatorname{arctanh} r/r = \sum_{j\geq 0} r^{2j}/(2j+1)$. Depolarization sends $(\mathbf{r}, \mathbf{v})$ to $(\lambda\mathbf{r}, \lambda\mathbf{v})$. For $|\lambda| \leq 1$ it consequently contracts the metric by at most λ^2. Integrating along the state segment and taking finite relative-entropy boundary limits proves the ordinary upper bound. Infinitesimal perturbations around $I_2/2$ attain the metric ratio λ^2, proving equality. In particular the proof covers the negative CP interval without any partial-transpose invariance claim.

This agrees with the ordinary contraction theory discussed in [7]; the purpose of the calculation is to make the three-way comparison in figure 2 self-contained.

C Finite checks and an excluded higher-order extension

C.1 A fully specified rational fixed-reference example

The source package contains small exact-arithmetic checks of examples, not a numerical substitute for the proofs above. Here is the fixed-reference example in full. Use ternary exclusion and a three-dimensional reference $R = \mathbb{C}^2 \oplus \mathbb{C}r$. Put

$$u_0 = \begin{pmatrix} 1/2 \\ 0 \end{pmatrix}, \quad u_1 = \begin{pmatrix} 0 \\ 1/2 \end{pmatrix}, \quad u_2 = \begin{pmatrix} -1/2 \\ -1/2 \end{pmatrix}, \quad n = 10000, \quad p = \frac{1}{1+n^2}, \quad q = 1-p, \quad t = \frac{1}{10}.$$

For a uniform bit $U = \pm 1$, define subnormalized reference blocks, indexed by the classical input x, by

$$\rho_x^{\pm} = \begin{pmatrix} q u_x u_x^* & \pm t\sqrt{pq}\, u_x \\ \pm t\sqrt{pq}\, u_x^* & p/3 \end{pmatrix}.$$

Every entry is rational since $\sqrt{pq} = n/(1+n^2)$. The states have trace one after summing over x because $\sum_x \|u_x\|^2 = 1$. On the support of u_x their Schur complements equal $p(1/3 - t^2) > 0$, proving positivity. Since $\sum_x u_x = 0$, the two reference marginals agree exactly; their input probabilities also agree. Thus $I(U;R) = I(U;X) = 0$. Their average deletes the displayed off-diagonal blocks. The output blocks of exclusion are $\rho'_y = (\sum_x \rho_x - \rho_y)/2$.

The supplied rational-interval calculation encloses the resulting ratio by

$$0.918490786084546654 < \frac{I(U;Y|R)}{I(U;X|R)} < 0.918490786084547090,$$

above the ordinary classical coefficient $1/2$. The input information is between $5.513760010993645 \times 10^{-9}$ and $5.513760010993648 \times 10^{-9}$ nats. This small signal is reported explicitly.

The certificate isolates the eigenvalues of the rational block matrices using rational characteristic-polynomial intervals. To bound logarithms it reduces to $1 \le z \le 2$ and uses, with $v = (z-1)/(z+1)$,

$$\log z = 2\sum_{j=0}^{N-1} \frac{v^{2j+1}}{2j+1} + R_N, \qquad 0 \le R_N \le \frac{2v^{2N+1}}{(2N+1)(1-v^2)}.$$

The remainder follows by bounding the positive tail termwise by a geometric series. Integer-square-root enclosures are used in the separate qubit-reference ternary example. All comparisons in these checks use rational endpoints rounded outward, not floating-point optimization.

The latter qubit-reference example has parameters $p = 1/10, s = 1/100, h = 1/100$ and states

$$\begin{aligned}\rho_{XR}(p,s) = {} & \frac{1-p}{2}|0\rangle\langle 0| \otimes |\psi_+\rangle\langle\psi_+| + \frac{1-p}{2}|1\rangle\langle 1| \otimes |\psi_-\rangle\langle\psi_-| \\ & + p|2\rangle\langle 2| \otimes |1\rangle\langle 1|, \\ |\psi_\pm\rangle = {} & \sqrt{1-s}|0\rangle \pm \sqrt{s}|1\rangle.\end{aligned}$$

A bit selects $p+h$ or $p-h$. Its conditional-information ratio lies between 0.927420641599022566 and 0.927420641599022567. Here the two reference marginals may differ; this example must not be identified with the exactly fixed-reference construction above.

C.2 An exact depolarizing quadratic check

For $\lambda = 1/2$ in dimension two, the matrix-unit Gram tuple with center $a_i = (5/8)\operatorname{Tr} F_i/2$ has defect

$$\frac{3}{8}\mathbf{G} - (\mathbf{A} - aI)(\mathbf{A} - aI)^*$$

with eigenvalues 0 twice, 3/32 four times and 15/128 twice. Let P_0 be its kernel projection, $C = P_0/2$ and $B = C(\mathbf{A} - aI)$. The included rational calculation verifies $\operatorname{Tr} C = 1$, $\operatorname{Tr}_A B = 0$, and the range conditions for the two quadratic forms. Their values are $q_C(B) = 3/8$ and $q_{C'}(B') = 9/64$, with pseudoinverses on these particular singular supports. Faithful mixtures of C give the same limiting ratio. The positive-center Choi calculation gives $M = 8/5$. These are exact finite checks of the value 3/8; the universal optimality proof is section 3–section 5.

C.3 Why an all-order Petz Rényi claim is false

There is no extension of (8.1) to every order above one. For qubit shrinkage 1/2 use the diagonal pair $P = (1,0)$, $Q = (9/10, 1/10)$. The input Petz order-ten divergence is $\log(10/9)$. The output laws are $(3/4, 1/4)$ and $(7/10, 3/10)$, so its output divergence is $(\log R)/9$, where the positive rational

$$R = \frac{(3/4)^{10}}{(7/10)^9} + \frac{(1/4)^{10}}{(3/10)^9}$$

satisfies $R^8 > (10/9)^{27}$ by direct integer arithmetic. Hence the output-to-input ratio exceeds 3/8. The source package includes the short rational check of this inequality. Order ten is outside the usual Petz data-processing range. This example does not settle the different question of exact coefficients for $1 < \alpha \le 2$.

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